

Physics Publications with Major Personal Contributions

Oliver Gutsche
August 01, 2026

1. Measurement of the fragmentation properties of jets containing $\Upsilon(nS)$ mesons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.
[arXiv:2607.25653](#) | 2026-07-28

2. Off-shell Higgs boson measurements: Yukawa couplings, self-coupling, compositeness, and width

Gevorgyan, Arzunik et al.
[arXiv:2607.23352](#) | 2026-07-25

3. Study of χ_c production in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Belyaev, Andrey et al.
[arXiv:2607.17847](#) | 2026-07-20

4. Evidence for sequential $\Upsilon(nS)$ suppression in light ion collisions

Belyaev, Andrey et al.
[arXiv:2607.13758](#) | 2026-07-15

5. Exploring the origin of D^0 meson elliptic flow in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV using event shape engineering

Gevorgyan, Arzunik et al.
[arXiv:2607.11017](#) | 2026-07-13

6. Search for soft unclustered energy patterns in proton-proton collisions at $\sqrt{s} = 13$ TeV using data scouting

Gevorgyan, Arzunik et al.
[arXiv:2607.09621](#) | 2026-07-10

7. Search for nonresonant triple Higgs boson production in the final state with six bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.
[arXiv:2607.05145](#) | 2026-07-06

8. Study of ZZ and ZH production in the $b\bar{b}\tau\tau$ final state and search for high-mass spin-0 and spin-1 resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.
[arXiv:2607.01589](#) | 2026-07-02

9. Search for physics beyond the standard model in four and three top quark production events using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2606.30862](#) | 2026-06-29

10. Precision luminosity measurement in proton-proton collisions at a center-of-mass energy of 13 TeV with the CMS detector at the Large Hadron Collider

Gevorgyan, Arzunik et al.
[arXiv:2606.26832](#) | 2026-06-25

11. Search for long-lived particles decaying into muons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV using data scouting

Gevorgyan, Arzunik et al.
[arXiv:2606.25840](#) | 2026-06-24

12. Observation of electroweak production of pairs of Z bosons in proton-proton collisions at 13 TeV

Gevorgyan, Arzunik et al.
[arXiv:2606.20558](#) | 2026-06-18

13. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and tau leptons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.
[arXiv:2606.03510](#) | 2026-06-02

14. Search for single production of a vector-like B' quark decaying to a top quark and a W boson in the single-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tauqeer, Komal et al.
[arXiv:2606.01423](#) | 2026-05-31

15. Dependence of two-particle azimuthal correlations on the forward rapidity gap width in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Hayrapetyan, Aram et al.
[arXiv:2606.00171](#) | 2026-05-29

16. Search for light scalar particles produced in Higgs boson decays in exclusive final states with two muons and two hadrons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2605.27875](#) | 2026-05-27

17. Search for a new heavy scalar resonance decaying to a pair of Z bosons in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2605.26462](#) | 2026-05-26

18. Search for low-mass resonances decaying to $\tau\tau$ and measurement of the $Y \rightarrow \tau\tau$ decay in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.
[arXiv:2605.25103](#) | 2026-05-24

19. Studies of $Z \rightarrow 4\ell$ decays in proton-proton collisions at $\sqrt{s} = 8$ and 13 TeV

Tumasyan, Armen et al.
[arXiv:2605.25158](#) | 2026-05-24

20. Search for Higgs boson decays into two neutral scalars with unequal masses in final states with b quarks and tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.
[arXiv:2605.19144](#) | 2026-05-18

21. Search for the nonresonant production of a pair of additional Higgs bosons in the Type-X two-Higgs-doublet model in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2605.16655](#) | 2026-05-15

22. Search for new physics in triple boson production in proton-proton collisions at $\sqrt{s} = 13$ TeV using the effective field theory approach

Hayrapetyan, Aram et al.
[arXiv:2605.15023](#) | 2026-05-14

23. Constraints on anomalous Higgs boson couplings to vector bosons and fermions using the $\gamma\gamma$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.14245](#) | 2026-05-14

24. First measurements of vector boson scattering in $W^{\pm}W^{\pm}$ and WZ production in all-leptonic final states at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2605.15396](#) | 2026-05-14

25. Search for pair production of additional neutral scalars within the Inert Doublet Model in a final state with two electrons or two muons in proton-proton collisions at $\sqrt{s} = 13$ TeV and 13.6 TeV

Gevorgyan, Arzunik et al.

[arXiv:2605.13614](#) | 2026-05-13

26. Search for single vector-like quark production in opposite-sign dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.13649](#) | 2026-05-13

27. Search for a new heavy scalar resonance decaying into the Higgs boson and a new scalar particle in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.02848](#) | 2026-05-04

28. Observation of the rare decay $\eta \rightarrow \mu^+ \mu^- e^+ e^-$

Hayrapetyan, Aram et al.

[arXiv:2605.00615](#) | 2026-05-01

29. Search for light charged Higgs bosons decaying to charm and strange quarks in $t\bar{t}$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.27795](#) | 2026-05-01

30. Improved results on Higgs boson pair production in the $4b$ final state

Hayrapetyan, Aram et al.

[arXiv:2604.27044](#) | 2026-04-29

31. Search for electroweakinos in compressed-spectrum scenarios with low-momentum isolated tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.25604](#) | 2026-04-28

32. Simultaneous measurements of N-subjettiness observables in jets from gluons and light-flavour quarks, and in decays of boosted W bosons and top quarks

Hayrapetyan, Aram et al.

[arXiv:2604.25538](#) | 2026-04-28

33. Measurement of the $Z \rightarrow \mu^+ \mu^-$ angular coefficients in pp collisions at $\sqrt{s} = 13$ TeV as functions of transverse momentum and rapidity

Gevorgyan, Arzunik et al.

[arXiv:2604.25678](#) | 2026-04-28

34. Search for associated production of a Higgs boson and two vector bosons via vector boson scattering at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.24531](#) | 2026-04-27

35. Search for dark matter produced in association with a dark Higgs boson decaying into a bottom quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.20999](#) | 2026-04-22

36. Search for the single production of vector-like quarks decaying into a W boson and a b quark using single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.17564](#) | 2026-04-19

37. Highly boosted dielectron identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.13320](#) | 2026-04-14

38. Search for heavy resonances decaying into four-lepton final states via light bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.14236](#) | 2026-04-14

39. A search for microscopic black holes, string balls, and sphalerons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10732](#) | 2026-04-12

40. Search for a new heavy resonance decaying to a top quark and a neutral scalar boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10729](#) | 2026-04-12

41. Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons

Hayrapetyan, Aram et al.

[arXiv:2604.09809](#) | 2026-04-10

42. Observation of nuclear suppression in coherent $\Upsilon(1S)$ photoproduction off heavy nuclei at the LHC

Belyaev, Andrey et al.

[arXiv:2604.05814](#) | 2026-04-07

43. Search for soft unclustered energy patterns produced in association with a W or Z boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 879 \(2026\) 140671](#) | [arXiv:2604.05996](#) | [DOI:10.1016/j.physletb.2026.140671](#) | 2026-04-07

44. Evidence of $ZZ\gamma$ production and observation of $4\ell\gamma$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.02594](#) | 2026-04-03

45. Search for Higgs boson pair production in the $b\bar{b}WW$ decay channel with two leptons in the final state using proton-proton collision data at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2604.02127](#) | 2026-04-02

46. Measurement of dijet angular distributions and search for beyond the standard model physics in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.
[arXiv:2603.25458](#) | 2026-03-26

47. Search for new particles decaying into top quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.
[arXiv:2603.23454](#) | 2026-03-24

48. Search for low-mass vector and scalar resonances decaying into a quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.21965](#) | 2026-03-23

49. Search for Higgs boson production at high transverse momentum in the WW decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.22233](#) | 2026-03-23

50. Measurement of the jet mass in hadronic decays of boosted W bosons at 13 TeV and extraction of the W boson mass

Hayrapetyan, Aram et al.
[JHEP 07 \(2026\) 054](#) | [arXiv:2603.19963](#) | [DOI:10.1007/JHEP07\(2026\)054](#) | 2026-03-20

51. Jet peak shapes based on two-particle angular correlations in lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.
[arXiv:2603.14385](#) | 2026-03-15

52. Measurement of the t-channel single top quark cross section in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.13592](#) | 2026-03-13

53. Search for Z' bosons decaying into charginos in final states with two oppositely charged leptons and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.11035](#) | 2026-03-11

54. Measurement of angular correlations inside jets induced by gluon polarization in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.03689](#) | 2026-03-04

55. Search for a narrow resonance with a mass between 10 and 70 GeV decaying to a pair of photons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.03250](#) | 2026-03-03

56. Combination of ATLAS and CMS searches for Higgs boson pair production at $\sqrt{s} = 13$ TeV

Aad, Georges et al.
[arXiv:2602.23991](#) | 2026-02-27

57. System-size dependence of charged-particle suppression in ultrarelativistic nucleus-nucleus collisions

Belyaev, Andrey et al.
[Phys.Lett.B 880 \(2026\) 140679](#) | [arXiv:2602.21325](#) | [DOI:10.1016/j.physletb.2026.140679](#) | 2026-02-24

58. Search for pair production of heavy resonances in final states with a photon and large-radius jets in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 112005](#) | [arXiv:2602.20477](#) | [DOI:10.1103/bwqs-vgqn](#) | 2026-02-24

59. Observation of the jet diffusion wake using dijets in heavy ion collisions

Hayrapetyan, Aram et al.

[arXiv:2602.19431](#) | 2026-02-23

60. Search for a new resonance decaying to a Higgs boson and a scalar boson in events with two b jets and two Z bosons in proton-proton collisions at $\sqrt{s} = 13.6 \text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2602.18223](#) | 2026-02-20

61. Combined measurements and interpretations of Higgs boson production and decay in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2602.18611](#) | 2026-02-20

62. First measurement of jet axis decorrelation with photon-tagged jets in pp and PbPb collisions at 5.02 TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18279](#) | 2026-02-20

63. Measurement of event shape variables using charged particles inside jets in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2602.17509](#) | 2026-02-19

64. Influence of strangeness on the anisotropic flow of prompt $D^{\pm}_s$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02 \text{ TeV}$

Belyaev, Andrey et al.

[arXiv:2602.14221](#) | 2026-02-15

65. High-level hadronic tau lepton triggers of the CMS experiment in proton-proton collisions at $\sqrt{s} = 13.6 \text{ TeV}$

Hayrapetyan, Aram et al.

[JINST 21 \(2026\) P04002](#) | [arXiv:2602.11359](#) | [DOI:10.1088/1748-0221/21/04/P04002](#) | 2026-02-11

66. Measurement of B meson production fraction ratios in proton-proton collisions at $\sqrt{s}=13\text{TeV}$ using open-charm and charmonium decays

Hayrapetyan, Aram et al.

[Phys.Rev.D 114 \(2026\) 012005](#) | [arXiv:2602.10270](#) | [DOI:10.1103/36cv-vst3](#) | 2026-02-10

67. Probing early parton emissions in heavy ion collisions using the Lund jet plane

Belyaev, Andrey et al.

[arXiv:2602.09271](#) | 2026-02-09

68. Observation of $\Upsilon(1S) + Z$ associated production and measurement of the effective double-parton scattering cross section in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2602.02770](#) | 2026-02-02

69. Observation of a family of all-charm tetraquarks

Hayrapetyan, Aram et al.

[arXiv:2602.02252](#) | 2026-02-02

70. Search for heavy scalar resonances decaying to Lorentz-boosted Higgs and Higgs-like bosons in the $b\bar{b}4q$ final state at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.00273](#) | 2026-01-30

71. Measurement and effective field theory interpretation of the photon-fusion production cross section of a pair of W bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

JHEP 06 (2026) 187 | [arXiv:2601.21574](#) | DOI:10.1007/JHEP06(2026)187 | 2026-01-29

72. Measurement of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ differential cross sections in pp collisions at $\sqrt{s}=13.6$ TeV

Hayrapetyan, Aram et al.

JHEP 07 (2026) 106 | [arXiv:2601.20023](#) | DOI:10.1007/JHEP07(2026)106 | 2026-01-27

73. Search for heavy resonances decaying into two Higgs bosons in the $b\bar{b}\tau^+\tau^-$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20011](#) | 2026-01-27

74. Search for heavy long-lived charged particles with level-1 trigger scouting data from proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

Phys.Lett.B 878 (2026) 140498 | [arXiv:2601.20063](#) | DOI:10.1016/j.physletb.2026.140498 | 2026-01-27

75. Strategy and performance of the CMS long-lived particle trigger program in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17544](#) | 2026-01-24

76. Search for the pair production of long-lived supersymmetric partners of the tau lepton in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

JHEP 07 (2026) 102 | [arXiv:2601.17576](#) | DOI:10.1007/JHEP07(2026)102 | 2026-01-24

77. Full event interpretation with machine-learning-based particle-flow reconstruction in the CMS detector

Hayrapetyan, Aram et al.

[arXiv:2601.17554](#) | 2026-01-24

78. Measurement of the $Z\gamma$ production cross section and search for anomalous neutral triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.14102](#) | 2026-01-20

79. Combination of searches for heavy vector boson resonances in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

JHEP 07 (2026) 005 | [arXiv:2601.12583](#) | DOI:10.1007/JHEP07(2026)005 | 2026-01-18

80. Search for dark matter produced in association with a Higgs boson decaying to bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

Phys.Rev.D 113 (2026) 092016 | [arXiv:2601.11330](#) | DOI:10.1103/2m87-3dlr | 2026-01-16

81. Search for a boosted Higgs boson decaying to bottom quark pairs in association with a W or Z boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 879 \(2026\) 140534](#) | [arXiv:2601.05362](#) | [DOI:10.1016/j.physletb.2026.140534](#) | 2026-01-08

82. Measurement of the Higgs boson total decay width using the $H \rightarrow WW \rightarrow e\nu\mu\nu$ decay channel in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 092014](#) | [arXiv:2601.05168](#) | [DOI:10.1103/rny9-zby1](#) | 2026-01-08

83. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A}$ with $\mathcal{A} \rightarrow \gamma\gamma$ in events with a semi-merged topology in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 07 \(2026\) 119](#) | [arXiv:2601.00183](#) | [DOI:10.1007/JHEP07\(2026\)119](#) | 2026-01-01

84. Search for charged Higgs bosons decaying into top and bottom quarks in lepton+jets final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2512.24471](#) | 2025-12-30

85. Machine-learning techniques for model-independent searches in dijet final states

Hayrapetyan, Aram et al.

[Mach.Learn.Sci.Tech. 7 \(2026\) 045008](#) | [arXiv:2512.20395](#) | [DOI:10.1088/2632-2153/ae7d87](#) | 2025-12-23

86. Characterization of the quantum state of top quark pairs produced in proton-proton collisions at $\sqrt{s}=13\text{TeV}$ using the beam and helicity bases

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) L051101](#) | [arXiv:2512.17557](#) | [DOI:10.1103/hkgw-dztk](#) | 2025-12-19

87. Measurement of $Z\gamma$ production in proton-proton collisions at $\sqrt{s} = 13.6\text{ TeV}$ and constraints on neutral triple gauge couplings

Hayrapetyan, Aram et al.

[arXiv:2512.08582](#) | 2025-12-09

88. Measurements of electroweak production of a photon in association with two jets in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 05 \(2026\) 020](#) | [arXiv:2512.00502](#) | [DOI:10.1007/JHEP05\(2026\)020](#) | 2025-11-29

89. Search for Light Pseudoscalar Bosons, Pair-Produced in Higgs Boson Decays in the Four-Electron Final State in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 181807](#) | [arXiv:2511.19563](#) | [DOI:10.1103/r53x-8fqp](#) | 2025-11-24

90. Search for new physics in the final state with a single photon and large missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 072020](#) | [arXiv:2511.17310](#) | [DOI:10.1103/f9r8-styj](#) | 2025-11-21

91. Search for Higgsinos in final states with low-momentum lepton-track pairs at 13 TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 092009](#) | [arXiv:2511.16394](#) | [DOI:10.1103/w3b5-92ls](#) | 2025-11-20

92. Search for heavy $H\gamma$ and $Z\gamma$ resonances with a bottom quark-antiquark pair in the final state in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2511.14583](#) | 2025-11-18

93. Search for low-mass hidden-valley dark showers with non-prompt muon pairs in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 189](#) | [arXiv:2511.11888](#) | [DOI:10.1007/JHEP03\(2026\)189](#) | 2025-11-14

94. Search for a new neutral gauge boson produced in association with one or two b jets and decaying into a pair of muons in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 07 \(2026\) 008](#) | [arXiv:2511.11853](#) | [DOI:10.1007/JHEP07\(2026\)008](#) | 2025-11-14

95. Search for long-lived particles using displaced vertices with low-momentum tracks in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2026\) 191](#) | [arXiv:2511.08212](#) | [DOI:10.1007/JHEP05\(2026\)191](#) | 2025-11-11

96. Identification of tau leptons using a convolutional neural network with domain adaptation

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P12032](#) | [arXiv:2511.05468](#) | [DOI:10.1088/1748-0221/20/12/P12032](#) | 2025-11-07

97. Jet fragmentation function and groomed substructure of bottom quark jets in proton-proton collisions at 5.02 TeV

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 147](#) | [arXiv:2511.10666](#) | [DOI:10.1007/JHEP04\(2026\)147](#) | 2025-11-06

98. Inclusive and differential measurements of the $t\bar{t}\gamma$ cross section and the $t\bar{t}\gamma / t\bar{t}$ cross section ratio in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2026\) 086](#) | [arXiv:2511.01995](#) | [DOI:10.1007/JHEP04\(2026\)086](#) | 2025-11-03

99. Characterizing the initial state and dynamical evolution in XeXe and PbPb collisions using multiparticle cumulants

Hayrapetyan, Aram et al.

[Phys.Lett.B 876 \(2026\) 140359](#) | [arXiv:2510.26766](#) | [DOI:10.1016/j.physletb.2026.140359](#) | 2025-10-30

100. Search for single production of a vector-like T quark decaying to a top quark and a neutral scalar boson in the lepton+jets final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 07 \(2026\) 041](#) | [arXiv:2510.25874](#) | [DOI:10.1007/JHEP07\(2026\)041](#) | 2025-10-29

101. Measurement of the ratio of the $B_c^+ \rightarrow J/\psi \tau^+ \nu_\tau$ and $B_c^+ \rightarrow J/\psi \mu^+ \nu_\mu$ branching fractions using three-prong τ lepton decays

Chekhovsky, Vladimir et al.

[Phys.Rev.D 113 \(2026\) L111101](#) | [arXiv:2510.21559](#) | [DOI:10.1103/mw5c-5rw3](#) | 2025-10-24

102. Search for dijet resonances with data scouting in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2026\) 133](#) | [arXiv:2510.21641](#) | [DOI:10.1007/JHEP04\(2026\)133](#) | 2025-10-24

103. Observation of tWZ Production at the CMS Experiment

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 081802](#) | [arXiv:2510.19080](#) | [DOI:10.1103/rk6w-1pcl](#) | 2025-10-21

104. Search for dark matter production in association with bottom quarks and a lepton pair in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 06 \(2026\) 014](#) | [arXiv:2510.12396](#) | [DOI:10.1007/JHEP06\(2026\)014](#) | 2025-10-14

105. Performance of heavy-flavour jet identification in Lorentz-boosted topologies in proton-proton collisions at $\sqrt{s} = 13$ TeV

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Hayrapetyan, Aram et al.

[Eur.Phys.J.C 86 \(2026\) 137](#) | [arXiv:2509.13635](#) | [DOI:10.1140/epjc/s10052-025-15176-9](#) | 2025-09-17

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Chekhovsky, Vladimir et al.

[Phys.Rev.Lett. 136 \(2026\) 122303](#) | [arXiv:2509.08626](#) | [DOI:10.1103/lckg-sdh9](#) | 2025-09-10

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[Phys.Rev.D 112 \(2025\) 112023](#) | [arXiv:2508.13900](#) | [DOI:10.1103/b26z-zmpy](#) | 2025-08-19

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[Phys.Rev.D 112 \(2025\) 112011](#) | [arXiv:2508.07512](#) | [DOI:10.1103/8lw4-6lyk](#) | 2025-08-11

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[JHEP 04 \(2025\) 115](#) | [arXiv:2412.02477](#) | [DOI:10.1007/JHEP04\(2025\)115](#) | 2024-12-03

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[JINST 20 \(2025\) P04012](#) | [arXiv:2411.19749](#) | [DOI:10.1088/1748-0221/20/04/P04012](#) | 2024-11-29

182. Constraints on standard model effective field theory for a Higgs boson produced in association with W or Z bosons in the $H \rightarrow b\bar{b}$ decay channel in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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183. Search for rare decays of the Z and Higgs bosons to a J/ψ or $\psi(2S)$ meson and a photon in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

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[Phys.Lett.B 864 \(2025\) 139406](#) | [arXiv:2411.11820](#) | [DOI:10.1016/j.physletb.2025.139406](#) | 2024-11-18

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186. Search for a heavy resonance decaying into a Z and a Higgs boson in events with an energetic jet and two electrons, two muons, or missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 04 \(2025\) 099](#) | [arXiv:2410.21631](#) | [DOI:10.1007/JHEP04\(2025\)099](#) | 2024-10-28

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Hayrapetyan, Aram et al.

[JHEP 06 \(2025\) 006](#) | [arXiv:2410.21137](#) | [DOI:10.1007/JHEP06\(2025\)006](#) | 2024-10-28

190. Search for pair production of heavy particles decaying to a top quark and a gluon in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 85 \(2025\) 342](#) | [arXiv:2410.20601](#) | [DOI:10.1140/epjc/s10052-024-13729-y](#) | 2024-10-27

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Hayrapetyan, Aram et al.

[Phys.Lett.B 862 \(2025\) 139296](#) | [arXiv:2410.18289](#) | [DOI:10.1016/j.physletb.2025.139296](#) | 2024-10-23

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[JINST 19 \(2024\) P11021](#) | [arXiv:2410.17038](#) | [DOI:10.1088/1748-0221/19/11/P11021](#) | 2024-10-22

193. Search for high-mass resonances in a final state comprising a gluon and two hadronically decaying W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 02 \(2025\) 199](#) | [arXiv:2410.17303](#) | [DOI:10.1007/JHEP02\(2025\)199](#) | 2024-10-22

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Tumasyan, Armen et al.

[arXiv:2410.04578](#) | 2024-10-06

196. Study of same-sign W boson scattering and anomalous couplings in events with one tau lepton from pp collisions at $\sqrt{s}=13$ TeV

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[Phys.Rev.D 111 \(2025\) 072003](#) | [arXiv:2409.17770](#) | [DOI:10.1103/PhysRevD.111.072003](#) | 2024-09-26

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[Phys.Rev.D 111 \(2025\) 092014](#) | [arXiv:2409.13663](#) | [DOI:10.1103/PhysRevD.111.092014](#) | 2024-09-20

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Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 112016](#) | [arXiv:2409.11067](#) | [DOI:10.1103/PhysRevD.110.112016](#) | 2024-09-17

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[Rept.Prog.Phys. 88 \(2025\) 037801](#) | [arXiv:2409.10806](#) | [DOI:10.1088/1361-6633/adaa13](#) | 2024-09-16

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[Phys.Lett.B 861 \(2025\) 139279](#) | [arXiv:2409.00838](#) | [DOI:10.1016/j.physletb.2025.139279](#) | 2024-09-01

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[JHEP 02 \(2025\) 050](#) | [arXiv:2408.17336](#) | [DOI:10.1007/JHEP02\(2025\)050](#) | 2024-08-30

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[Phys.Lett.B 866 \(2025\) 139526](#) | [arXiv:2408.07622](#) | [DOI:10.1016/j.physletb.2025.139526](#) | 2024-08-14

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Hayrapetyan, Aram et al.

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[Phys.Rev.D 111 \(2025\) L051102](#) | [arXiv:2408.00678](#) | [DOI:10.1103/PhysRevD.111.L051102](#) | 2024-08-01

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[JHEP 12 \(2024\) 172](#) | [arXiv:2407.20425](#) | [DOI:10.1007/JHEP12\(2024\)172](#) | 2024-07-29

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[Phys.Lett.B 861 \(2025\) 139231](#) | [arXiv:2406.05101](#) | [DOI:10.1016/j.physletb.2024.139231](#) | 2024-06-07

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[Rept.Prog.Phys. 87 \(2024\) 117801](#) | [arXiv:2406.03976](#) | [DOI:10.1088/1361-6633/ad7e4d](#) | 2024-06-06

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[Phys.Rept. 1115 \(2025\) 3](#) | [arXiv:2405.18661](#) | [DOI:10.1016/j.physrep.2024.11.005](#) | 2024-05-28

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[Phys.Lett.B 860 \(2025\) 139067](#) | [arXiv:2405.18149](#) | [DOI:10.1016/j.physletb.2024.139067](#) | 2024-05-28

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[Phys.Lett.B 860 \(2025\) 139202](#) | [arXiv:2405.16566](#) | [DOI:10.1016/j.physletb.2024.139202](#) | 2024-05-26

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[Phys.Lett.B 857 \(2024\) 138979](#) | [arXiv:2405.14757](#) | [DOI:10.1016/j.physletb.2024.138979](#) | 2024-05-23

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[Phys.Rept. 1115 \(2025\) 2367](#) | [arXiv:2405.10785](#) | [DOI:10.1016/j.physrep.2024.11.007](#) | 2024-05-17

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[JHEP 08 \(2024\) 215](#) | [arXiv:2405.09320](#) | [DOI:10.1007/JHEP08\(2024\)215](#) | 2024-05-15

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[Phys.Lett.B 861 \(2025\) 139088](#) | [arXiv:2405.02737](#) | [DOI:10.1016/j.physletb.2024.139088](#) | 2024-05-04

236. Search for New Resonances Decaying to Pairs of Merged Diphotons in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.Lett. 134 \(2025\) 041801](#) | [arXiv:2405.00834](#) | [DOI:10.1103/PhysRevLett.134.041801](#) | 2024-05-01

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[Phys.Rev.Lett. 133 \(2024\) 161805](#) | [arXiv:2404.18298](#) | [DOI:10.1103/PhysRevLett.133.161805](#) | 2024-04-28

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[JINST 19 \(2024\) P09012](#) | [arXiv:2404.17377](#) | [DOI:10.1088/1748-0221/19/09/P09012](#) | 2024-04-26

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[Eur.Phys.J.C 84 \(2024\) 842](#) | [arXiv:2404.16082](#) | [DOI:10.1140/epjc/s10052-024-13116-7](#) | 2024-04-24

240. Search for Higgs boson pair production with one associated vector boson in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

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[Comput.Softw.Big Sci. 8 \(2024\) 19](#) | [arXiv:2404.06614](#) | [DOI:10.1007/s41781-024-00121-4](#) | 2024-04-09

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243. Searches for Pair-Produced Multijet Resonances Using Data Scouting in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.Lett. 133 \(2024\) 201803](#) | [arXiv:2404.02992](#) | [DOI:10.1103/PhysRevLett.133.201803](#) | 2024-04-03

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[Phys.Lett.B 857 \(2024\) 138964](#) | [arXiv:2403.20201](#) | [DOI:10.1016/j.physletb.2024.138964](#) | 2024-03-29

245. Search for ZZ and ZH production in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

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[Eur.Phys.J.C 84 \(2024\) 712](#) | [arXiv:2403.20241](#) | [DOI:10.1140/epjc/s10052-024-13021-z](#) | 2024-03-29

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[Phys.Rept. 1115 \(2025\) 2342](#) | [arXiv:2403.16926](#) | [DOI:10.1016/j.physrep.2024.09.004](#) | 2024-03-25

247. Enriching the physics program of the CMS experiment via data scouting and data parking

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[Phys.Rept. 1115 \(2025\) 2344](#) | [arXiv:2403.16134](#) | [DOI:10.1016/j.physrep.2024.09.006](#) | 2024-03-24

248. Performance of the CMS electromagnetic calorimeter in pp collisions at $\sqrt{s} = 13$ TeV

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[JINST 19 \(2024\) P09004](#) | [arXiv:2403.15518](#) | [DOI:10.1088/1748-0221/19/09/P09004](#) | 2024-03-22

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[Phys.Rev.D 109 \(2024\) L111101](#) | [arXiv:2403.11352](#) | [DOI:10.1103/PhysRevD.109.L111101](#) | 2024-03-17

250. Search for the decay of the Higgs boson to a pair of light pseudoscalar bosons in the final state with four bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 06 \(2024\) 097](#) | [arXiv:2403.10341](#) | [DOI:10.1007/JHEP06\(2024\)097](#) | 2024-03-15

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252. Search for Soft Unclustered Energy Patterns in Proton-Proton Collisions at 13 TeV

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[Phys.Rev.Lett. 133 \(2024\) 191902](#) | [arXiv:2403.05311](#) | [DOI:10.1103/PhysRevLett.133.191902](#) | 2024-03-08

253. Search for long-lived heavy neutrinos in the decays of B mesons produced in proton-proton collisions at $\sqrt{s} = 13$ TeV

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254. Search for dark QCD with emerging jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 07 \(2024\) 142](#) | [arXiv:2403.01556](#) | [DOI:10.1007/JHEP07\(2024\)142](#) | 2024-03-03

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[Phys.Rept. 1115 \(2025\) 2369](#) | [arXiv:2403.01313](#) | [DOI:10.1016/j.physrep.2024.12.002](#) | 2024-03-02

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[Eur.Phys.J.C 84 \(2024\) 779](#) | [arXiv:2403.00657](#) | [DOI:10.1140/epjc/s10052-024-12925-0](#) | 2024-03-01

257. Search for heavy neutral leptons in final states with electrons, muons, and hadronically decaying tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 06 \(2024\) 123](#) | [arXiv:2403.00100](#) | [DOI:10.1007/JHEP06\(2024\)123](#) | 2024-02-29

258. Search for long-lived heavy neutral leptons decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 110 \(2024\) 012004](#) | [arXiv:2402.18658](#) | [DOI:10.1103/PhysRevD.110.012004](#) | 2024-02-28

259. Search for baryon number violation in top quark production and decay using proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 132 \(2024\) 241802](#) | [arXiv:2402.18461](#) | [DOI:10.1103/PhysRevLett.132.241802](#) | 2024-02-28

260. Observation of the $\Xi_b^- \rightarrow \psi(2S)\Xi^-$ decay and studies of the $\Xi_b(5945)^0$ baryon in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 110 \(2024\) 012002](#) | [arXiv:2402.17738](#) | [DOI:10.1103/PhysRevD.110.012002](#) | 2024-02-27

261. Search for long-lived particles using displaced vertices and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112005](#) | [arXiv:2402.15804](#) | [DOI:10.1103/PhysRevD.109.112005](#) | 2024-02-24

262. Portable Acceleration of CMS Computing Workflows with Coprocessors as a Service

Hayrapetyan, Aram et al.

[Comput.Softw.Big Sci. 8 \(2024\) 17](#) | [arXiv:2402.15366](#) | [DOI:10.1007/s41781-024-00124-1](#) | 2024-02-23

263. Search for long-lived particles decaying to final states with a pair of muons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

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[JHEP 05 \(2024\) 047](#) | [arXiv:2402.14491](#) | [DOI:10.1007/JHEP05\(2024\)047](#) | 2024-02-22

264. Measurement of Energy Correlators inside Jets and Determination of the Strong Coupling $\alpha_S(m_Z)$

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[Phys.Rev.Lett. 133 \(2024\) 071903](#) | [arXiv:2402.13864](#) | [DOI:10.1103/PhysRevLett.133.071903](#) | 2024-02-21

265. Search for bottom-type vectorlike quark pair production in dileptonic and fully hadronic final states in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 052004](#) | [arXiv:2402.13808](#) | [DOI:10.1103/PhysRevD.110.052004](#) | 2024-02-21

266. Search for exotic decays of the Higgs boson to a pair of pseudoscalars in the $\mu\mu b\bar{b}$ and $\tau\tau b\bar{b}$ final states

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[Eur.Phys.J.C 84 \(2024\) 493](#) | [arXiv:2402.13358](#) | [DOI:10.1140/epjc/s10052-024-12727-4](#) | 2024-02-20

267. Search for a scalar or pseudoscalar dilepton resonance produced in association with a massive vector boson or top quark-antiquark pair in multilepton events at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 110 \(2024\) 012013](#) | [arXiv:2402.11098](#) | [DOI:10.1103/PhysRevD.110.012013](#) | 2024-02-16

268. Search for Fractionally Charged Particles in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 134 \(2025\) 131802](#) | [arXiv:2402.09932](#) | [DOI:10.1103/PhysRevLett.134.131802](#) | 2024-02-15

269. Combination of Measurements of the Top Quark Mass from Data Collected by the ATLAS and CMS Experiments at $\sqrt{s}=7$ and 8 TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 261902](#) | [arXiv:2402.08713](#) | [DOI:10.1103/PhysRevLett.132.261902](#) | 2024-02-13

270. Differential cross section measurements for the production of top quark pairs and of additional jets using dilepton events from pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2025\) 064](#) | [arXiv:2402.08486](#) | [DOI:10.1007/JHEP02\(2025\)064](#) | 2024-02-13

271. Search for pair production of scalar and vector leptoquarks decaying to muons and bottom quarks in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112003](#) | [arXiv:2402.08668](#) | [DOI:10.1103/PhysRevD.109.112003](#) | 2024-02-09

272. Search for long-lived particles decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 032007](#) | [arXiv:2402.01898](#) | [DOI:10.1103/PhysRevD.110.032007](#) | 2024-02-02

273. Combined search for electroweak production of winos, binos, higgsinos, and sleptons in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112001](#) | [arXiv:2402.01888](#) | [DOI:10.1103/PhysRevD.109.112001](#) | 2024-02-02

274. Observation of the $\Lambda_b^0 \rightarrow J/\psi \Xi^- K^+$ decay

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 1062](#) | [arXiv:2401.16303](#) | [DOI:10.1140/epjc/s10052-024-13114-9](#) | 2024-01-29

275. Nonresonant central exclusive production of charged-hadron pairs in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112013](#) | [arXiv:2401.14494](#) | [DOI:10.1103/PhysRevD.109.112013](#) | 2024-01-25

276. Measurement of the double-differential inclusive jet cross section in proton-proton collisions at $\sqrt{s} = 5.02 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 01 \(2025\) 011](#) | [arXiv:2401.11355](#) | [DOI:10.1007/JHEP01\(2025\)011](#) | 2024-01-20

277. Test of lepton flavor universality in $B^{\pm} \rightarrow K^{\pm} \mu^+ \mu^-$ and $B^{\pm} \rightarrow K^{\pm} e^+ e^-$ decays in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 87 \(2024\) 077802](#) | [arXiv:2401.07090](#) | [DOI:10.1088/1361-6633/ad4e65](#) | 2024-01-13

278. Extracting the speed of sound in quark-gluon plasma with ultrarelativistic lead-lead collisions at the LHC

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 87 \(2024\) 077801](#) | [arXiv:2401.06896](#) | [DOI:10.1088/1361-6633/ad4b9b](#) | 2024-01-12

279. Observation of Enhanced Long-Range Elliptic Anisotropies Inside High-Multiplicity Jets in pp Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 142301](#) | [arXiv:2312.17103](#) | [DOI:10.1103/PhysRevLett.133.142301](#) | 2023-12-28

280. Elliptic anisotropy measurement of the $f_0(980)$ hadron in proton-lead collisions and evidence for its quark-antiquark composition

Hayrapetyan, Aram et al.

[Nature Commun. 16 \(2025\) 7990](#) | [arXiv:2312.17092](#) | [DOI:10.1038/s41467-025-56200-6](#) | 2023-12-28

281. Measurement of multidifferential cross sections for dijet production in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 85 \(2025\) 72](#) | [arXiv:2312.16669](#) | [DOI:10.1140/epjc/s10052-024-13606-8](#) | 2023-12-27

282. Measurement of the primary Lund jet plane density in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 116](#) | [arXiv:2312.16343](#) | [DOI:10.1007/JHEP05\(2024\)116](#) | 2023-12-26

283. Evidence for tWZ production in proton-proton collisions at $\sqrt{s}=13$ TeV in multilepton final states

Hayrapetyan, Aram et al.

[Phys.Lett.B 855 \(2024\) 138815](#) | [arXiv:2312.11668](#) | [DOI:10.1016/j.physletb.2024.138815](#) | 2023-12-18

284. Search for Long-Lived Heavy Neutral Leptons with Lepton Flavour Conserving or Violating Decays to a Jet and a Charged Lepton

Hayrapetyan, Aram et al.

[JHEP 03 \(2024\) 105](#) | [arXiv:2312.07484](#) | [DOI:10.1007/JHEP03\(2024\)105](#) | 2023-12-12

285. Search for flavor changing neutral current interactions of the top quark in final states with a photon and additional jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072004](#) | [arXiv:2312.08229](#) | [DOI:10.1103/PhysRevD.109.072004](#) | 2023-12-11

286. Measurement of simplified template cross sections of the Higgs boson produced in association with W or Z bosons in the $H \rightarrow b\bar{b}$ decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 109 \(2024\) 092011](#) | [arXiv:2312.07562](#) | [DOI:10.1103/PhysRevD.109.092011](#) | 2023-12-09

287. Search for charged-lepton flavor violation in the production and decay of top quarks using trilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) 012009](#) | [arXiv:2312.03199](#) | [DOI:10.1103/PhysRevD.111.012009](#) | 2023-12-05

288. Search for the lepton flavor violating $\tau \rightarrow 3\mu$ decay in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 853 \(2024\) 138633](#) | [arXiv:2312.02371](#) | [DOI:10.1016/j.physletb.2024.138633](#) | 2023-12-04

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Tumasyan, Armen et al.

[JHEP 2024 \(2024\) 106](#) | [arXiv:2311.11370](#) | [DOI:10.1007/JHEP02\(2024\)106](#) | 2023-11-19

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[Phys.Lett.B 850 \(2024\) 138478](#) | [arXiv:2311.03261](#) | [DOI:10.1016/j.physletb.2024.138478](#) | 2023-11-06

291. Search for high-mass exclusive diphoton production with tagged protons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 110 \(2024\) 012010](#) | [arXiv:2311.02725](#) | [DOI:10.1103/PhysRevD.110.012010](#) | 2023-11-05

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Hayrapetyan, Aram et al.

[Phys.Lett.B 852 \(2024\) 138582](#) | [arXiv:2311.00130](#) | [DOI:10.1016/j.physletb.2024.138582](#) | 2023-10-31

293. Search for W' bosons decaying to a top and a bottom quark in leptonic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 046](#) | [arXiv:2310.19893](#) | [DOI:10.1007/JHEP05\(2024\)046](#) | 2023-10-30

294. Search for Narrow Trijet Resonances in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 011801](#) | [arXiv:2310.14023](#) | [DOI:10.1103/PhysRevLett.133.011801](#) | 2023-10-21

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[JHEP 03 \(2024\) 134](#) | [arXiv:2310.12229](#) | [DOI:10.1007/JHEP03\(2024\)134](#) | 2023-10-18

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Tumasyan, Armen et al.

[JHEP 06 \(2024\) 187](#) | [arXiv:2310.11231](#) | [DOI:10.1007/JHEP06\(2024\)187](#) | 2023-10-17

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[Phys.Rev.Lett. 132 \(2024\) 121901](#) | [arXiv:2310.05164](#) | [DOI:10.1103/PhysRevLett.132.121901](#) | 2023-10-08

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Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P02031](#) | [arXiv:2310.03844](#) | [DOI:10.1088/1748-0221/19/02/P02031](#) | 2023-10-05

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[Phys.Lett.B 850 \(2024\) 138518](#) | [arXiv:2310.03233](#) | [DOI:10.1016/j.physletb.2024.138518](#) | 2023-10-04

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Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112009](#) | [arXiv:2310.03154](#) | [DOI:10.1103/PhysRevD.109.112009](#) | 2023-10-04

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[JHEP 05 \(2024\) 316](#) | [arXiv:2310.01643](#) | [DOI:10.1007/JHEP05\(2024\)316](#) | 2023-10-02

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Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072007](#) | [arXiv:2309.16823](#) | [DOI:10.1103/PhysRevD.109.072007](#) | 2023-09-28

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[JHEP 12 \(2023\) 070](#) | [arXiv:2309.16003](#) | [DOI:10.1007/JHEP12\(2023\)070](#) | 2023-09-27

304. Inclusive and differential cross section measurements of $t\bar{t}b\bar{b}$ production in the lepton+jets channel at $\sqrt{s} = 13$ TeV

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[JHEP 05 \(2024\) 042](#) | [arXiv:2309.14442](#) | [DOI:10.1007/JHEP05\(2024\)042](#) | 2023-09-25

305. Measurement of the τ lepton polarization in Z boson decays in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 101](#) | [arXiv:2309.12408](#) | [DOI:10.1007/JHEP01\(2024\)101](#) | 2023-09-21

306. Development of the CMS detector for the CERN LHC Run 3

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[JINST 19 \(2024\) P05064](#) | [arXiv:2309.05466](#) | [DOI:10.1088/1748-0221/19/05/P05064](#) | 2023-09-11

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[Phys.Rev.Lett. 132 \(2024\) 021803](#) | [arXiv:2309.03501](#) | [DOI:10.1103/PhysRevLett.132.021803](#) | 2023-09-07

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Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 26](#) | [arXiv:2309.01008](#) | [DOI:10.1140/epjc/s10052-023-12268-2](#) | 2023-09-02

309. Search for a third-generation leptoquark coupled to a τ lepton and a b quark through single, pair, and nonresonant production in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 05 \(2024\) 311](#) | [arXiv:2308.07826](#) | [DOI:10.1007/JHEP05\(2024\)311](#) | 2023-08-15

310. Search for Scalar Leptoquarks Produced via τ -Lepton-Quark Scattering in pp Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 061801](#) | [arXiv:2308.06143](#) | [DOI:10.1103/PhysRevLett.132.061801](#) | 2023-08-11

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[Eur.Phys.J.C 84 \(2024\) 27](#) | [arXiv:2308.02285](#) | [DOI:10.1140/epjc/s10052-023-12258-4](#) | 2023-08-04

312. Measurement of the Higgs boson production via vector boson fusion and its decay into bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 173](#) | [arXiv:2308.01253](#) | [DOI:10.1007/JHEP01\(2024\)173](#) | 2023-08-02

313. Search for new physics in multijet events with at least one photon and large missing transverse momentum in proton-proton collisions at 13 TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 046](#) | [arXiv:2307.16216](#) | [DOI:10.1007/JHEP10\(2023\)046](#) | 2023-07-30

314. Search for physics beyond the standard model in top quark production with additional leptons in the context of effective field theory

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 068](#) | [arXiv:2307.15761](#) | [DOI:10.1007/JHEP12\(2023\)068](#) | 2023-07-28

315. Study of charm hadronization with prompt Λ_c^+ baryons in proton-proton and lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 01 \(2024\) 128](#) | [arXiv:2307.11186](#) | [DOI:10.1007/JHEP01\(2024\)128](#) | 2023-07-20

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Tumasyan, Armen et al.

[JHEP 08 \(2024\) 148](#) | [arXiv:2307.11185](#) | [DOI:10.1007/JHEP08\(2024\)148](#) | 2023-07-20

317. Search for a high-mass dimuon resonance produced in association with b quark jets at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 043](#) | [arXiv:2307.08708](#) | [DOI:10.1007/JHEP10\(2023\)043](#) | 2023-07-14

318. Search for Z' bosons decaying to pairs of heavy Majorana neutrinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 11 \(2023\) 181](#) | [arXiv:2307.06959](#) | [DOI:10.1007/JHEP11\(2023\)181](#) | 2023-07-12

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[Phys.Rev.C 109 \(2024\) 024914](#) | [arXiv:2306.11574](#) | [DOI:10.1103/PhysRevC.109.024914](#) | 2023-06-20

320. Performance of the local reconstruction algorithms for the CMS hadron calorimeter with Run 2 data

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[JINST 18 \(2023\) P11017](#) | [arXiv:2306.10355](#) | [DOI:10.1088/1748-0221/18/11/P11017](#) | 2023-06-17

321. New Structures in the $J/\psi J/\psi$ Mass Spectrum in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 111901](#) | [arXiv:2306.07164](#) | [DOI:10.1103/PhysRevLett.132.111901](#) | 2023-06-12

322. Search for the lepton-flavor violating decay of the Higgs boson and additional Higgs bosons in the $e\mu$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 108 \(2023\) 072004](#) | [arXiv:2305.18106](#) | [DOI:10.1103/PhysRevD.108.072004](#) | 2023-05-29

323. Measurements of the azimuthal anisotropy of prompt and nonprompt charmonia in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 10 \(2023\) 115](#) | [arXiv:2305.16928](#) | [DOI:10.1007/JHEP10\(2023\)115](#) | 2023-05-26

324. Observation of four top quark production in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 847 \(2023\) 138290](#) | [arXiv:2305.13439](#) | [DOI:10.1016/j.physletb.2023.138290](#) | 2023-05-22

325. Search for Inelastic Dark Matter in Events with Two Displaced Muons and Missing Transverse Momentum in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 041802](#) | [arXiv:2305.11649](#) | [DOI:10.1103/PhysRevLett.132.041802](#) | 2023-05-19

326. Search for resonances in events with photon and jet final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2023\) 189](#) | [arXiv:2305.07998](#) | [DOI:10.1007/JHEP12\(2023\)189](#) | 2023-05-13

327. Measurements of inclusive and differential cross sections for the Higgs boson production and decay to four-leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 08 \(2023\) 040](#) | [arXiv:2305.07532](#) | [DOI:10.1007/JHEP08\(2023\)040](#) | 2023-05-12

328. Observation of the rare decay of the η meson to four muons

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 131 \(2023\) 091903](#) | [arXiv:2305.04904](#) | [DOI:10.1103/PhysRevLett.131.091903](#) | 2023-05-08

329. Search for top squark pair production in a final state with at least one hadronically decaying tau lepton in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 110](#) | [arXiv:2304.07174](#) | [DOI:10.1007/JHEP07\(2023\)110](#) | 2023-04-14

330. Probing Small Bjorken- x Nuclear Gluonic Structure via Coherent J/ψ Photoproduction in Ultraperipheral Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 262301](#) | [arXiv:2303.16984](#) | [DOI:10.1103/PhysRevLett.131.262301](#) | 2023-03-29

331. Observation of the $\chi(3S)$ Meson and Suppression of χ States in Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 133 \(2024\) 022302](#) | [arXiv:2303.17026](#) | [DOI:10.1103/PhysRevLett.133.022302](#) | 2023-03-29

332. First measurement of the top quark pair production cross section in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Tumasyan, Armen et al.

[JHEP 08 \(2023\) 204](#) | [arXiv:2303.10680](#) | [DOI:10.1007/JHEP08\(2023\)204](#) | 2023-03-19

333. A search for new physics in central exclusive production using the missing mass technique with the CMS detector and the CMS-TOTEM precision proton spectrometer

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 827](#) | [arXiv:2303.04596](#) | [DOI:10.1140/epjc/s10052-023-11687-5](#) | 2023-03-08

334. Evidence for Four-Top Quark Production in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 844 \(2023\) 138076](#) | [arXiv:2303.03864](#) | [DOI:10.1016/j.physletb.2023.138076](#) | 2023-03-07

335. A search for decays of the Higgs boson to invisible particles in events with a top-antitop quark pair or a vector boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 933](#) | [arXiv:2303.01214](#) | [DOI:10.1140/epjc/s10052-023-11952-7](#) | 2023-03-02

336. Search for a vector-like quark $T' \rightarrow tH$ via the diphoton decay mode of the Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 057](#) | [arXiv:2302.12802](#) | [DOI:10.1007/JHEP09\(2023\)057](#) | 2023-02-24

337. Measurement of the top quark mass using a profile likelihood approach with the lepton + jets final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 963](#) | [arXiv:2302.01967](#) | [DOI:10.1140/epjc/s10052-023-12050-4](#) | 2023-02-03

338. Search for top squarks in the four-body decay mode with single lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 060](#) | [arXiv:2301.08096](#) | [DOI:10.1007/JHEP06\(2023\)060](#) | 2023-01-19

339. First Measurement of the Forward Rapidity Gap Distribution in pPb Collisions at $\sqrt{s_{NN}}=8.16$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 092004](#) | [arXiv:2301.07630](#) | [DOI:10.1103/PhysRevD.108.092004](#) | 2023-01-18

340. K_S^0 and $\Lambda(\Lambda^0)$ two-particle femtoscopic correlations in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 857 \(2024\) 138936](#) | [arXiv:2301.05290](#) | [DOI:10.1016/j.physletb.2024.138936](#) | 2023-01-12

341. Search for new physics in the τ lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[JHEP 09 \(2023\) 051](#) | [arXiv:2212.12604](#) | [DOI:10.1007/JHEP09\(2023\)051](#) | 2022-12-23

342. Measurement of the electroweak production of $W\gamma$ in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032017](#) | [arXiv:2212.12592](#) | [DOI:10.1103/PhysRevD.108.032017](#) | 2022-12-23

343. Measurement of the $B^0_S \rightarrow \mu^+ \mu^-$ decay properties and search for the $B^0 \rightarrow \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137955](#) | [arXiv:2212.10311](#) | [DOI:10.1016/j.physletb.2023.137955](#) | 2022-12-20

344. Search for long-lived particles using out-of-time trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 210](#) | [arXiv:2212.06695](#) | [DOI:10.1007/JHEP07\(2023\)210](#) | 2022-12-13

345. Measurement of the dependence of the hadron production fraction ratio f_s / f_u and f_d / f_u on B meson kinematic variables in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 121901](#) | [arXiv:2212.02309](#) | [DOI:10.1103/PhysRevLett.131.121901](#) | 2022-12-05

346. Measurements of azimuthal anisotropy of nonprompt D0 mesons in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 850 \(2024\) 138389](#) | [arXiv:2212.01636](#) | [DOI:10.1016/j.physletb.2023.138389](#) | 2022-12-03

347. Search for high-mass exclusive $\gamma\gamma \rightarrow WW$ and $\gamma\gamma \rightarrow ZZ$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 229](#) | [arXiv:2211.16320](#) | [DOI:10.1007/JHEP07\(2023\)229](#) | 2022-11-29

348. Search for Higgs Boson and Observation of Z Boson through their Decay into a Charm Quark-Antiquark Pair in Boosted Topologies in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041801](#) | [arXiv:2211.14181](#) | [DOI:10.1103/PhysRevLett.131.041801](#) | 2022-11-25

349. Search for supersymmetry in final states with a single electron or muon using angular correlations and heavy-object identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 149](#) | [arXiv:2211.08476](#) | [DOI:10.1007/JHEP09\(2023\)149](#) | 2022-11-15

350. Measurement of the differential $t\bar{t}$ production cross section as a function of the jet mass and extraction of the top quark mass in hadronic decays of boosted top quarks

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 560](#) | [arXiv:2211.01456](#) | [DOI:10.1140/epjc/s10052-023-11587-8](#) | 2022-11-02

351. Azimuthal correlations in Z +jets events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 722](#) | [arXiv:2210.16139](#) | [DOI:10.1140/epjc/s10052-023-11833-z](#) | 2022-10-28

352. Measurements of jet multiplicity and jet transverse momentum in multijet events in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 742](#) | [arXiv:2210.13557](#) | [DOI:10.1140/epjc/s10052-023-11753-y](#) | 2022-10-24

353. Application of Portable Parallelization Strategies for GPUs on track reconstruction kernels

Kwok, Martin et al.

[J.Phys.Conf.Ser. 3206 \(2026\) 012052](#) | [DOI:10.1088/1742-6596/3206/1/012052](#) | 2026

354. Search for medium effects using jets from bottom quarks in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137849](#) | [arXiv:2210.08547](#) | [DOI:10.1016/j.physletb.2023.137849](#) | 2022-10-16

355. Azimuthal anisotropy of dijet events in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 139](#) | [arXiv:2210.08325](#) | [DOI:10.1007/JHEP07\(2023\)139](#) | 2022-10-15

356. Proton reconstruction with the CMS-TOTEM Precision Proton Spectrometer

Tumasyan, Armen et al.

[JINST 18 \(2023\) P09009](#) | [arXiv:2210.05854](#) | [DOI:10.1088/1748-0221/18/09/P09009](#) | 2022-10-11

357. Search for a heavy composite Majorana neutrino in events with dilepton signatures from proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 843 \(2023\) 137803](#) | [arXiv:2210.03082](#) | [DOI:10.1016/j.physletb.2023.137803](#) | 2022-10-06

358. Search for new heavy resonances decaying to WW, WZ, ZZ, WH, or ZH boson pairs in the all-jets final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137813](#) | [arXiv:2210.00043](#) | [DOI:10.1016/j.physletb.2023.137813](#) | 2022-09-30

359. Search for pair production of vector-like quarks in leptonic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 020](#) | [arXiv:2209.07327](#) | [DOI:10.1007/JHEP07\(2023\)020](#) | 2022-09-15

360. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A} \rightarrow 4\gamma$ with events containing two merged diphotons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 101801](#) | [arXiv:2209.06197](#) | [DOI:10.1103/PhysRevLett.131.101801](#) | 2022-09-13

361. Search for new physics using effective field theory in 13 TeV pp collision events that contain a top quark pair and a boosted Z or Higgs boson

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032008](#) | [arXiv:2208.12837](#) | [DOI:10.1103/PhysRevD.108.032008](#) | 2022-08-26

362. Measurement of the Higgs boson inclusive and differential fiducial production cross sections in the diphoton decay channel with pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 091](#) | [arXiv:2208.12279](#) | [DOI:10.1007/JHEP07\(2023\)091](#) | 2022-08-25

363. Search for pair-produced vector-like leptons in final states with third-generation leptons and at least three b quark jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137713](#) | [arXiv:2208.09700](#) | [DOI:10.1016/j.physletb.2023.137713](#) | 2022-08-20

364. Measurement of the cross section of top quark-antiquark pair production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 219](#) | [arXiv:2208.06485](#) | [DOI:10.1007/JHEP07\(2023\)219](#) | 2022-08-12

365. Search for CP violation in $t\bar{t}H$ and tH production in multilepton channels in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 092](#) | [arXiv:2208.02686](#) | [DOI:10.1007/JHEP07\(2023\)092](#) | 2022-08-04

366. Measurement of the $t\bar{t}$ charge asymmetry in events with highly Lorentz-boosted top quarks in pp collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137703](#) | [arXiv:2208.02751](#) | [DOI:10.1016/j.physletb.2023.137703](#) | 2022-08-04

367. Searches for additional Higgs bosons and for vector leptoquarks in $\tau\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 073](#) | [arXiv:2208.02717](#) | [DOI:10.1007/JHEP07\(2023\)073](#) | 2022-08-04

368. Search for the exotic decay of the Higgs boson into two light pseudoscalars with four photons in the final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 148](#) | [arXiv:2208.01469](#) | [DOI:10.1007/JHEP07\(2023\)148](#) | 2022-08-02

369. Measurement of inclusive and differential cross sections for single top quark production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 046](#) | [arXiv:2208.00924](#) | [DOI:10.1007/JHEP07\(2023\)046](#) | 2022-08-01

370. Search for the Higgs boson decay to a pair of electrons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137783](#) | [arXiv:2208.00265](#) | [DOI:10.1016/j.physletb.2023.137783](#) | 2022-07-30

371. Search for new physics in the lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 067](#) | [arXiv:2202.06075](#) | [DOI:10.1007/JHEP07\(2022\)067](#) | 2022-07-11

372. Measurement of the top quark pole mass using $t\bar{t}$ +jet events in the dilepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 077](#) | [arXiv:2207.02270](#) | [DOI:10.1007/JHEP07\(2023\)077](#) | 2022-07-05

373. Search for direct pair production of supersymmetric partners of τ leptons in the final state with two hadronically decaying τ leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 012011](#) | [arXiv:2207.02254](#) | [DOI:10.1103/PhysRevD.108.012011](#) | 2022-07-05

374. Search for a charged Higgs boson decaying into a heavy neutral Higgs boson and a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 032](#) | [arXiv:2207.01046](#) | [DOI:10.1007/JHEP09\(2023\)032](#) | 2022-07-03

375. Search for Higgs boson pairs decaying to WW^*WW^* , $WW^*\tau\tau$, and $\tau\tau\tau\tau$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 095](#) | [arXiv:2206.10268](#) | [DOI:10.1007/JHEP07\(2023\)095](#) | 2022-06-21

376. Search for nonresonant Higgs boson pair production in the four leptons plus twob jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 130](#) | [arXiv:2206.10657](#) | [DOI:10.1007/JHEP06\(2023\)130](#) | 2022-06-21

377. Search for resonant and nonresonant production of pairs of dijet resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 161](#) | [arXiv:2206.09997](#) | [DOI:10.1007/JHEP07\(2023\)161](#) | 2022-06-20

378. Search for nonresonant Higgs boson pair production in final state with two bottom quarks and two tau leptons in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137531](#) | [arXiv:2206.09401](#) | [DOI:10.1016/j.physletb.2022.137531](#) | 2022-06-19

379. Measurements of the Higgs boson production cross section and couplings in the W boson pair decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 667](#) | [arXiv:2206.09466](#) | [DOI:10.1140/epjc/s10052-023-11632-6](#) | 2022-06-19

380. Probing Heavy Majorana Neutrinos and the Weinberg Operator through Vector Boson Fusion Processes in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 011803](#) | [arXiv:2206.08956](#) | [DOI:10.1103/PhysRevLett.131.011803](#) | 2022-06-17

381. Precision measurement of the Z boson invisible width in pp collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137563](#) | [arXiv:2206.07110](#) | [DOI:10.1016/j.physletb.2022.137563](#) | 2022-06-14

382. Observation of τ lepton pair production in ultraperipheral lead-lead collisions at $\sqrt{s_{\text{NN}}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 151803](#) | [arXiv:2206.05192](#) | [DOI:10.1103/PhysRevLett.131.151803](#) | 2022-06-10

383. Search for Higgs boson decays into Z and J/ψ and for Higgs and Z boson decays into J/ψ or Y pairs in pp collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137534](#) | [arXiv:2206.03525](#) | [DOI:10.1016/j.physletb.2022.137534](#) | 2022-06-07

384. Observation of same-sign WW production from double parton scattering in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 091803](#) | [arXiv:2206.02681](#) | [DOI:10.1103/PhysRevLett.131.091803](#) | 2022-06-06

385. Combination of inclusive top-quark pair production cross-section measurements using ATLAS and CMS data at $\sqrt{s} = 7$ and 8 TeV

Aad, G. et al.

[JHEP 07 \(2023\) 213](#) | [arXiv:2205.13830](#) | [DOI:10.1007/JHEP07\(2023\)213](#) | 2022-05-27

386. Search for electroweak production of charginos and neutralinos at $\sqrt{s}=13$ TeV in final states containing hadronic decays of WW, WZ, or WH and missing transverse momentum

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137460](#) | [arXiv:2205.09597](#) | [DOI:10.1016/j.physletb.2022.137460](#) | 2022-05-19

387. Search for long-lived particles decaying to a pair of muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 228](#) | [arXiv:2205.08582](#) | [DOI:10.1007/JHEP05\(2023\)228](#) | 2022-05-17

388. Search for CP violating top quark couplings in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 023](#) | [arXiv:2205.07434](#) | [DOI:10.1007/JHEP07\(2023\)023](#) | 2022-05-15

389. Search for heavy resonances and quantum black holes in $e\mu$, $e\tau$, and $\mu\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 227](#) | [arXiv:2205.06709](#) | [DOI:10.1007/JHEP05\(2023\)227](#) | 2022-05-13

390. Search for Nonresonant Pair Production of Highly Energetic Higgs Bosons Decaying to Bottom Quarks

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041803](#) | [arXiv:2205.06667](#) | [DOI:10.1103/PhysRevLett.131.041803](#) | 2022-05-13

391. Search for Higgs Boson Decay to a Charm Quark-Antiquark Pair in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 061801](#) | [arXiv:2205.05550](#) | [DOI:10.1103/PhysRevLett.131.061801](#) | 2022-05-11

392. Observation of electroweak $W+W^-$ pair production in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 841 \(2023\) 137495](#) | [arXiv:2205.05711](#) | [DOI:10.1016/j.physletb.2022.137495](#) | 2022-05-11

393. Measurement of the mass dependence of the transverse momentum of lepton pairs in Drell-Yan production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 628](#) | [arXiv:2205.04897](#) | [DOI:10.1140/epjc/s10052-023-11631-7](#) | 2022-05-10

394. Constraints on anomalous Higgs boson couplings to vector bosons and fermions from the production of Higgs bosons using the $\tau\tau$ final state

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032013](#) | [arXiv:2205.05120](#) | [DOI:10.1103/PhysRevD.108.032013](#) | 2022-05-10

395. CMS pythia 8 colour reconnection tunes based on underlying-event data

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 587](#) | [arXiv:2205.02905](#) | [DOI:10.1140/epjc/s10052-023-11630-8](#) | 2022-05-05

396. Measurement of differential cross sections for the production of a Z boson in association with jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 052004](#) | [arXiv:2205.02872](#) | [DOI:10.1103/PhysRevD.108.052004](#) | 2022-05-05

397. Search for CP violation using tt events in the lepton+jets channel in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 081](#) | [arXiv:2205.02314](#) | [DOI:10.1007/JHEP06\(2023\)081](#) | 2022-05-04

398. Search for narrow resonances in the b-tagged dijet mass spectrum in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 012009](#) | [arXiv:2205.01835](#) | [DOI:10.1103/PhysRevD.108.012009](#) | 2022-05-03

399. Strange hadron collectivity in pPb and PbPb collisions

Tumasyan, Armen et al.

400. Azimuthal Correlations within Exclusive Dijets with Large Momentum Transfer in Photon-Lead Collisions

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 051901](#) | [arXiv:2205.00045](#) | [DOI:10.1103/PhysRevLett.131.051901](#) | 2022-04-29

401. Two-particle azimuthal correlations in γp interactions using pPb collisions at $\sqrt{s_{NN}}=8.16$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137905](#) | [arXiv:2204.13486](#) | [DOI:10.1016/j.physletb.2023.137905](#) | 2022-04-28

402. Search for light Higgs bosons from supersymmetric cascade decays in pp collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 571](#) | [arXiv:2204.13532](#) | [DOI:10.1140/epjc/s10052-023-11581-0](#) | 2022-04-28

403. Search for Higgs boson decays to a Z boson and a photon in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 233](#) | [arXiv:2204.12945](#) | [DOI:10.1007/JHEP05\(2023\)233](#) | 2022-04-27

404. Measurements of Higgs boson production in the decay channel with a pair of τ leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 562](#) | [arXiv:2204.12957](#) | [DOI:10.1140/epjc/s10052-023-11452-8](#) | 2022-04-27

405. Search for a massive scalar resonance decaying to a light scalar and a Higgs boson in the four b quarks final state with boosted topology

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137392](#) | [arXiv:2204.12413](#) | [DOI:10.1016/j.physletb.2022.137392](#) | 2022-04-26

406. Reconstruction of decays to merged photons using end-to-end deep learning with domain continuation in the CMS detector

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 052002](#) | [arXiv:2204.12313](#) | [DOI:10.1103/PhysRevD.108.052002](#) | 2022-04-26

407. Search for new particles in an extended Higgs sector with four b quarks in the final state at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 835 \(2022\) 137566](#) | [arXiv:2203.00480](#) | [DOI:10.1016/j.physletb.2022.137566](#) | 2022-03-01

408. Search for a W' boson decaying to a vector-like quark and a top or bottom quark in the all-jets final state at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2022\) 088](#) | [arXiv:2202.12988](#) | [DOI:10.1007/JHEP09\(2022\)088](#) | 2022-02-25

409. Measurement of the Drell-Yan forward-backward asymmetry at high dilepton masses in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 2022 \(2022\) 063](#) | [arXiv:2202.12327](#) | [DOI:10.1007/JHEP08\(2022\)063](#) | 2022-02-24

410. Nuclear modification of Υ states in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 835 \(2022\) 137397](#) | [arXiv:2202.11807](#) | [DOI:10.1016/j.physletb.2022.137397](#) | 2022-02-23

411. Search for Higgs Boson Pair Production in the Four b Quark Final State in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 081802](#) | [arXiv:2202.09617](#) | [DOI:10.1103/PhysRevLett.129.081802](#) | 2022-02-19

412. Inclusive nonresonant multilepton probes of new phenomena at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 112007](#) | [arXiv:2202.08676](#) | [DOI:10.1103/PhysRevD.105.112007](#) | 2022-02-17

413. Measurement of the Higgs boson width and evidence of its off-shell contributions to ZZ production

Tumasyan, Armen et al.

[Nature Phys. 18 \(2022\) 1329](#) | [arXiv:2202.06923](#) | [DOI:10.1038/s41567-022-01682-0](#) | 2022-02-14

414. Search for invisible decays of the Higgs boson produced via vector boson fusion in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 092007](#) | [arXiv:2201.11585](#) | [DOI:10.1103/PhysRevD.105.092007](#) | 2022-01-27

415. Observation of $B^0 \rightarrow \psi(2S) K^0_S \pi^+ \pi^-$ and $B^0 \rightarrow \psi(2S) K^0_S$ decays

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 499](#) | [arXiv:2201.09131](#) | [DOI:10.1140/epjc/s10052-022-10315-y](#) | 2022-01-22

416. Search for resonances decaying to three W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 021802](#) | [arXiv:2201.08476](#) | [DOI:10.1103/PhysRevLett.129.021802](#) | 2022-01-20

417. Identification of hadronic tau lepton decays using a deep neural network

Tumasyan, Armen et al.

[JINST 17 \(2022\) P07023](#) | [arXiv:2201.08458](#) | [DOI:10.1088/1748-0221/17/07/P07023](#) | 2022-01-20

418. Precision measurement of the W boson decay branching fractions in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 072008](#) | [arXiv:2201.07861](#) | [DOI:10.1103/PhysRevD.105.072008](#) | 2022-01-19

419. Search for charged-lepton flavor violation in top quark production and decay in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 082](#) | [arXiv:2201.07859](#) | [DOI:10.1007/JHEP06\(2022\)082](#) | 2022-01-19

420. Measurement of the inclusive and differential $t\bar{t}\gamma$ cross sections in the dilepton channel and effective field theory interpretation in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 091](#) | [arXiv:2201.07301](#) | [DOI:10.1007/JHEP05\(2022\)091](#) | 2022-01-18

421. Search for long-lived heavy neutral leptons with displaced vertices in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 081](#) | [arXiv:2201.05578](#) | [DOI:10.1007/JHEP07\(2022\)081](#) | 2022-01-14

422. Search for higgsinos decaying to two Higgs bosons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 014](#) | [arXiv:2201.04206](#) | [DOI:10.1007/JHEP05\(2022\)014](#) | 2022-01-11

423. Observation of the B_c^+ Meson in Pb-Pb and pp Collisions at $\sqrt{s_{NN}}=5.02$ TeV and Measurement of its Nuclear Modification Factor

Tumasyan, Armen et al.

[Phys.Rev.Lett. 128 \(2022\) 252301](#) | [arXiv:2201.02659](#) | [DOI:10.1103/PhysRevLett.128.252301](#) | 2022-01-07

424. Search for high-mass resonances decaying to a jet and a Lorentz-boosted resonance in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 832 \(2022\) 137263](#) | [arXiv:2201.02140](#) | [DOI:10.1016/j.physletb.2022.137263](#) | 2022-01-06

425. Search for single production of a vector-like T quark decaying to a top quark and a Z boson in the final state with jets and missing transverse momentum at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 093](#) | [arXiv:2201.02227](#) | [DOI:10.1007/JHEP05\(2022\)093](#) | 2022-01-06

426. Search for long-lived particles decaying into muon pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV collected with a dedicated high-rate data stream

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 062](#) | [arXiv:2112.13769](#) | [DOI:10.1007/JHEP04\(2022\)062](#) | 2021-12-27

427. Search for resonances decaying to three W bosons in the hadronic final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012002](#) | [arXiv:2112.13090](#) | [DOI:10.1103/PhysRevD.106.012002](#) | 2021-12-24

428. Probing Charm Quark Dynamics via Multiparticle Correlations in Pb-Pb Collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 022001](#) | [arXiv:2112.12236](#) | [DOI:10.1103/PhysRevLett.129.022001](#) | 2021-12-22

429. Search for resonant production of strongly coupled dark matter in proton-proton collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 156](#) | [arXiv:2112.11125](#) | [DOI:10.1007/JHEP06\(2022\)156](#) | 2021-12-21

430. Search for flavor-changing neutral current interactions of the top quark and the Higgs boson decaying to a bottom quark-antiquark pair at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 169](#) | [arXiv:2112.09734](#) | [DOI:10.1007/JHEP02\(2022\)169](#) | 2021-12-17

431. Measurement of the production cross section for Z+b jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 092014](#) | [arXiv:2112.09659](#) | [DOI:10.1103/PhysRevD.105.092014](#) | 2021-12-17

432. Measurement of the inclusive tt production cross section in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 144](#) | [arXiv:2112.09114](#) | [DOI:10.1007/JHEP04\(2022\)144](#) | 2021-12-16

433. Evidence for WW/WZ vector boson scattering in the decay channel $\ell\nu qq$ produced in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 834 \(2022\) 137438](#) | [arXiv:2112.05259](#) | [DOI:10.1016/j.physletb.2022.137438](#) | 2021-12-09

434. Search for a right-handed W boson and a heavy neutrino in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 047](#) | [arXiv:2112.03949](#) | [DOI:10.1007/JHEP04\(2022\)047](#) | 2021-12-07

435. Search for heavy resonances decaying to a pair of Lorentz-boosted Higgs bosons in final states with leptons and a bottom quark pair at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 005](#) | [arXiv:2112.03161](#) | [DOI:10.1007/JHEP05\(2022\)005](#) | 2021-12-06

436. Measurements of the associated production of a W boson and a charm quark in proton-proton collisions at $\sqrt{s}=8$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 1094](#) | [arXiv:2112.00895](#) | [DOI:10.1140/epjc/s10052-022-10897-7](#) | 2021-12-01

437. Measurement of $W^{\pm}\gamma$ differential cross sections in proton-proton collisions at $\sqrt{s}=13$ TeV and effective field theory constraints

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 052003](#) | [arXiv:2111.13948](#) | [DOI:10.1103/PhysRevD.105.052003](#) | 2021-11-27

438. Search for heavy resonances decaying to ZZ or ZW and axion-like particles mediating nonresonant ZZ or ZH production at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 087](#) | [arXiv:2111.13669](#) | [DOI:10.1007/JHEP04\(2022\)087](#) | 2021-11-26

439. Search for a heavy resonance decaying into a top quark and a W boson in the lepton+jets final state at $\sqrt{s}=13$ TeV

Tumasyan, A. et al.

[JHEP 04 \(2022\) 048](#) | [arXiv:2111.10216](#) | [DOI:10.1007/JHEP04\(2022\)048](#) | 2021-11-19

440. Measurement and QCD analysis of double-differential inclusive jet cross sections in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 142](#) | [arXiv:2111.10431](#) | [DOI:10.1007/JHEP02\(2022\)142](#) | 2021-11-19

441. Strategies and performance of the CMS silicon tracker alignment during LHC Run 2

Tumasyan, Armen et al.

[Nucl.Instrum.Meth.A 1037 \(2022\) 166795](#) | [arXiv:2111.08757](#) | [DOI:10.1016/j.nima.2022.166795](#) | 2021-11-16

442. Search for supersymmetry in final states with two or three soft leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 091](#) | [arXiv:2111.06296](#) | [DOI:10.1007/JHEP04\(2022\)091](#) | 2021-11-11

443. Observation of triple J/ψ meson production in proton-proton collisions

Tumasyan, Armen et al.

[Nature Phys. 19 \(2023\) 338](#) | [arXiv:2111.05370](#) | [DOI:10.1038/s41567-022-01838-y](#) | 2021-11-09

444. Study of dijet events with large rapidity separation in proton-proton collisions at $\sqrt{s}=2.76$ TeV

Tumasyan, Armen et al.

[JHEP 03 \(2022\) 189](#) | [arXiv:2111.04605](#) | [DOI:10.1007/JHEP03\(2022\)189](#) | 2021-11-08

445. A new calibration method for charm jet identification validated with proton-proton collision events at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[JINST 17 \(2022\) P03014](#) | [arXiv:2111.03027](#) | [DOI:10.1088/1748-0221/17/03/P03014](#) | 2021-11-04

446. Inclusive and differential cross section measurements of single top quark production in association with a Z boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 107](#) | [arXiv:2111.02860](#) | [DOI:10.1007/JHEP02\(2022\)107](#) | 2021-11-04

447. Search for Flavor-Changing Neutral Current Interactions of the Top Quark and Higgs Boson in Final States with Two Photons in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, A. et al.

[Phys.Rev.Lett. 129 \(2022\) 032001](#) | [arXiv:2111.02219](#) | [DOI:10.1103/PhysRevLett.129.032001](#) | 2021-11-03

448. Search for low-mass dilepton resonances in Higgs boson decays to four-lepton final states in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Tumasyan, A. et al.

[Eur.Phys.J.C 82 \(2022\) 290](#) | [arXiv:2111.01299](#) | [DOI:10.1140/epjc/s10052-022-10127-0](#) | 2021-11-01

449. Search for long-lived particles produced in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 03 \(2022\) 160](#) | [arXiv:2110.13218](#) | [DOI:10.1007/JHEP03\(2022\)160](#) | 2021-10-25

450. Measurement of the inclusive and differential WZ production cross sections, polarization angles, and triple gauge couplings in pp collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 032](#) | [arXiv:2110.11231](#) | [DOI:10.1007/JHEP07\(2022\)032](#) | 2021-10-21

451. First Search for Exclusive Diphoton Production at High Mass with Tagged Protons in Proton-Proton Collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 011801](#) | [arXiv:2110.05916](#) | [DOI:10.1103/PhysRevLett.129.011801](#) | 2021-10-12

452. Search for long-lived particles decaying to leptons with large impact parameter in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 153](#) | [arXiv:2110.04809](#) | [DOI:10.1140/epjc/s10052-022-10027-3](#) | 2021-10-10

453. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and τ leptons in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 012](#) | [arXiv:2110.04836](#) | [DOI:10.1007/JHEP06\(2022\)012](#) | 2021-10-10

454. Measurement of double-parton scattering in inclusive production of four jets with low transverse momentum in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 177](#) | [arXiv:2109.13822](#) | [DOI:10.1007/JHEP01\(2022\)177](#) | 2021-09-28

455. Search for heavy resonances decaying to $Z(\nu\bar{\nu})V(q\bar{q}')$ in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012004](#) | [arXiv:2109.08268](#) | [DOI:10.1103/PhysRevD.106.012004](#) | 2021-09-17

456. Search for heavy resonances decaying to WW, WZ, or WH boson pairs in the lepton plus merged jet final state in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 032008](#) | [arXiv:2109.06055](#) | [DOI:10.1103/PhysRevD.105.032008](#) | 2021-09-13

457. Study of quark and gluon jet substructure in Z+jet and dijet events from pp collisions

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 188](#) | [arXiv:2109.03340](#) | [DOI:10.1007/JHEP01\(2022\)188](#) | 2021-09-07

458. Observation of B^0_s mesons and measurement of the B^0_s/B^+ yield ratio in PbPb collisions at $\sqrt{s_{NN}} = 5.02\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 829 \(2022\) 137062](#) | [arXiv:2109.01908](#) | [DOI:10.1016/j.physletb.2022.137062](#) | 2021-09-04

459. Observation of $t\bar{W}$ production in the single-lepton channel in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 111](#) | [arXiv:2109.01706](#) | [DOI:10.1007/JHEP11\(2021\)111](#) | 2021-09-03

460. Measurement of the top quark mass using events with a single reconstructed top quark in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 161](#) | [arXiv:2108.10407](#) | [DOI:10.1007/JHEP12\(2021\)161](#) | 2021-08-23

461. Measurement of differential $t\bar{t}$ production cross sections in the full kinematic range using lepton+jets events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 092013](#) | [arXiv:2108.02803](#) | [DOI:10.1103/PhysRevD.104.092013](#) | 2021-08-05

462. Probing effective field theory operators in the associated production of top quarks with a Z boson in multilepton final states at $\sqrt{s} = 13$ TeV

Lee, Kyeongpil et al.

[JHEP 12 \(2021\) 083](#) | [arXiv:2107.13896](#) | [DOI:10.1007/JHEP12\(2021\)083](#) | 2021-07-29

463. Search for new particles in events with energetic jets and large missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 153](#) | [arXiv:2107.13021](#) | [DOI:10.1007/JHEP11\(2021\)153](#) | 2021-07-27

464. Search for chargino-neutralino production in events with Higgs and W bosons using 137 fb^{-1} of proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 045](#) | [arXiv:2107.12553](#) | [DOI:10.1007/JHEP10\(2021\)045](#) | 2021-07-27

465. Measurement of the inclusive and differential Higgs boson production cross sections in the decay mode to a pair of τ leptons in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 128 \(2022\) 081805](#) | [arXiv:2107.11486](#) | [DOI:10.1103/PhysRevLett.128.081805](#) | 2021-07-23

466. Combined searches for the production of supersymmetric top quark partners in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 81 \(2021\) 970](#) | [arXiv:2107.10892](#) | [DOI:10.1140/epjc/s10052-021-09721-5](#) | 2021-07-22

467. Search for Long-Lived Particles Decaying in the CMS End Cap Muon Detectors in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 261804](#) | [arXiv:2107.04838](#) | [DOI:10.1103/PhysRevLett.127.261804](#) | 2021-07-10

468. Measurement of the inclusive and differential $t\bar{t}\gamma$ cross sections in the single-lepton channel and EFT interpretation at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 180](#) | [arXiv:2107.01508](#) | [DOI:10.1007/JHEP12\(2021\)180](#) | 2021-07-03

469. Measurement of prompt open-charm production cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 225](#) | [arXiv:2107.01476](#) | [DOI:10.1007/JHEP11\(2021\)225](#) | 2021-07-03

470. Measurements of the electroweak diboson production cross sections in proton-proton collisions at $\sqrt{s} = 5.02$ TeV using leptonic decays

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 191801](#) | [arXiv:2107.01137](#) | [DOI:10.1103/PhysRevLett.127.191801](#) | 2021-07-02

471. Search for electroweak production of charginos and neutralinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 147](#) | [arXiv:2106.14246](#) | [DOI:10.1007/JHEP04\(2022\)147](#) | 2021-06-27

472. Fragmentation of jets containing a prompt J/ψ meson in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 825 \(2022\) 136842](#) | [arXiv:2106.13235](#) | [DOI:10.1016/j.physletb.2021.136842](#) | 2021-06-23

473. Measurement of the electroweak production of $Z\gamma$ and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 072001](#) | [arXiv:2106.11082](#) | [DOI:10.1103/PhysRevD.104.072001](#) | 2021-06-21

474. Search for $W\gamma$ resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV using hadronic decays of Lorentz-boosted W bosons

Tumasyan, Armen et al.

[Phys.Lett.B 826 \(2022\) 136888](#) | [arXiv:2106.10509](#) | [DOI:10.1016/j.physletb.2022.136888](#) | 2021-06-19

475. Search for a heavy Higgs boson decaying into two lighter Higgs bosons in the $\tau\tau b\bar{b}$ final state at 13 TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 057](#) | [arXiv:2106.10361](#) | [DOI:10.1007/JHEP11\(2021\)057](#) | 2021-06-18

476. Study of Z boson plus jets events using variables sensitive to double-parton scattering in pp collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 176](#) | [arXiv:2105.14511](#) | [DOI:10.1007/JHEP10\(2021\)176](#) | 2021-05-30

477. Measurements of the $pp \rightarrow W^{\pm}\gamma\gamma$ and $pp \rightarrow Z\gamma\gamma$ cross sections at $\sqrt{s} = 13$ TeV and limits on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 174](#) | [arXiv:2105.12780](#) | [DOI:10.1007/JHEP10\(2021\)174](#) | 2021-05-26

478. Search for strongly interacting massive particles generating trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 213](#) | [arXiv:2105.09178](#) | [DOI:10.1140/epjc/s10052-022-10095-5](#) | 2021-05-19

479. Search for lepton-flavor violating decays of the Higgs boson in the $\mu\tau$ and $e\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032013](#) | [arXiv:2105.03007](#) | [DOI:10.1103/PhysRevD.104.032013](#) | 2021-05-06

480. Search for long-lived particles decaying to jets with displaced vertices in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[Phys.Rev.D 104 \(2021\) 052011](#) | [arXiv:2104.13474](#) | [DOI:10.1103/PhysRevD.104.052011](#) | 2021-04-27

481. Search for a heavy resonance decaying to a top quark and a W boson at $\sqrt{s} = 13$ TeV in the fully hadronic final state

Sirunyan, Albert M et al.

[JHEP 12 \(2021\) 106](#) | [arXiv:2104.12853](#) | [DOI:10.1007/JHEP12\(2021\)106](#) | 2021-04-26

482. Constraints on anomalous Higgs boson couplings to vector bosons and fermions in its production and decay using the four-lepton final state

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052004](#) | [arXiv:2104.12152](#) | [DOI:10.1103/PhysRevD.104.052004](#) | 2021-04-25

483. Search for W' bosons decaying to a top and a bottom quark at $\sqrt{s}=13\text{TeV}$ in the hadronic final state

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136535](#) | [arXiv:2104.04831](#) | [DOI:10.1016/j.physletb.2021.136535](#) | 2021-04-10

484. Search for charged Higgs bosons produced in vector boson fusion processes and decaying into vector boson pairs in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 723](#) | [arXiv:2104.04762](#) | [DOI:10.1140/epjc/s10052-021-09472-3](#) | 2021-04-10

485. Precision luminosity measurement in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$ in 2015 and 2016 at CMS

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 800](#) | [arXiv:2104.01927](#) | [DOI:10.1140/epjc/s10052-021-09538-2](#) | 2021-04-05

486. Constraints on the Initial State of Pb-Pb Collisions via Measurements of Z-Boson Yields and Azimuthal Anisotropy at $\sqrt{s_{\text{NN}}}=5.02\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 127 \(2021\) 102002](#) | [arXiv:2103.14089](#) | [DOI:10.1103/PhysRevLett.127.102002](#) | 2021-03-25

487. Measurements of Higgs boson production cross sections and couplings in the diphoton decay channel at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 027](#) | [arXiv:2103.06956](#) | [DOI:10.1007/JHEP07\(2021\)027](#) | 2021-03-11

488. Measurements of production cross sections of the Higgs boson in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 488](#) | [arXiv:2103.04956](#) | [DOI:10.1140/epjc/s10052-021-09200-x](#) | 2021-03-08

489. Using Z Boson Events to Study Parton-Medium Interactions in Pb-Pb Collisions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 128 \(2022\) 122301](#) | [arXiv:2103.04377](#) | [DOI:10.1103/PhysRevLett.128.122301](#) | 2021-03-07

490. Search for resonant and nonresonant new phenomena in high-mass dilepton final states at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 208](#) | [arXiv:2103.02708](#) | [DOI:10.1007/JHEP07\(2021\)208](#) | 2021-03-03

491. Search for top squark production in fully-hadronic final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052001](#) | [arXiv:2103.01290](#) | [DOI:10.1103/PhysRevD.104.052001](#) | 2021-03-01

492. Study of Drell-Yan dimuon production in proton-lead collisions at $\sqrt{s_{\text{NN}}} = 8.16\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 182](#) | [arXiv:2102.13648](#) | [DOI:10.1007/JHEP05\(2021\)182](#) | 2021-02-26

493. First measurement of large area jet transverse momentum spectra in heavy-ion collisions

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 284](#) | [arXiv:2102.13080](#) | [DOI:10.1007/JHEP05\(2021\)284](#) | 2021-02-25

494. Evidence for X(3872) in Pb-Pb Collisions and Studies of its Prompt Production at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 128 \(2022\) 032001](#) | [arXiv:2102.13048](#) | [DOI:10.1103/PhysRevLett.128.032001](#) | 2021-02-25

495. Measurements of angular distance and momentum ratio distributions in three-jet and Z + two-jet final states in pp collisions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 852](#) | [arXiv:2102.08816](#) | [DOI:10.1140/epjc/s10052-021-09570-2](#) | 2021-02-17

496. Search for a heavy vector resonance decaying to a Z^0 boson and a Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 688](#) | [arXiv:2102.08198](#) | [DOI:10.1140/epjc/s10052-021-09348-6](#) | 2021-02-16

497. Search for top squarks in final states with two top quarks and several light-flavor jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032006](#) | [arXiv:2102.06976](#) | [DOI:10.1103/PhysRevD.104.032006](#) | 2021-02-13

498. Hard color-singlet exchange in dijet events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032009](#) | [arXiv:2102.06945](#) | [DOI:10.1103/PhysRevD.104.032009](#) | 2021-02-13

499. Performance of the CMS muon trigger system in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JINST 16 \(2021\) P07001](#) | [arXiv:2102.04790](#) | [DOI:10.1088/1748-0221/16/07/P07001](#) | 2021-02-09

500. Observation of a New Excited Beauty Strange Baryon Decaying to $\Xi^0_{b1} \pi^0$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 126 \(2021\) 252003](#) | [arXiv:2102.04524](#) | [DOI:10.1103/PhysRevLett.126.252003](#) | 2021-02-08

501. Measurement of the $W\gamma$ Production Cross Section in Proton-Proton Collisions at $\sqrt{s}=13$ TeV and Constraints on Effective Field Theory Coefficients

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 126 \(2021\) 252002](#) | [arXiv:2102.02283](#) | [DOI:10.1103/PhysRevLett.126.252002](#) | 2021-02-03

502. Measurements of the differential cross sections of the production of Z + jets and γ + jets and of Z boson emission collinear with a jet in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 285](#) | [arXiv:2102.02238](#) | [DOI:10.1007/JHEP05\(2021\)285](#) | 2021-02-03

503. In-medium modification of dijets in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 116](#) | [arXiv:2101.04720](#) | [DOI:10.1007/JHEP05\(2021\)116](#) | 2021-01-12

504. Measurement of the Z boson differential production cross section using its invisible decay mode ($Z\nu\bar{\nu}$) in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 205](#) | [arXiv:2012.09254](#) | [DOI:10.1007/JHEP05\(2021\)205](#) | 2020-12-16

505. First measurement of the cross section for top quark pair production with additional charm jets using dileptonic final states in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136565](#) | [arXiv:2012.09225](#) | [DOI:10.1016/j.physletb.2021.136565](#) | 2020-12-16

506. Search for supersymmetry in final states with two oppositely charged same-flavor leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 123](#) | [arXiv:2012.08600](#) | [DOI:10.1007/JHEP04\(2021\)123](#) | 2020-12-15

507. Electron and photon reconstruction and identification with the CMS experiment at the CERN LHC

Sirunyan, Albert M et al.

[JINST 16 \(2021\) P05014](#) | [arXiv:2012.06888](#) | [DOI:10.1088/1748-0221/16/05/P05014](#) | 2020-12-12

508. Measurement of differential cross sections for Z bosons produced in association with charm jets in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 109](#) | [arXiv:2012.04119](#) | [DOI:10.1007/JHEP04\(2021\)109](#) | 2020-12-07

509. Search for new physics in top quark production with additional leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV using effective field theory

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[Phys.Rev.D 100 \(2019\) 112002](#) | [arXiv:1903.06973](#) | [DOI:10.1103/PhysRevD.100.112002](#) | 2019-03-16

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Sirunyan, Albert M et al.

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[Phys.Lett.B 792 \(2019\) 369](#) | [arXiv:1812.06504](#) | [DOI:10.1016/j.physletb.2019.03.059](#) | 2018-12-16

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678. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state with two muons and two b quarks in pp collisions at 13 TeV

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Sirunyan, Albert M et al.

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[JHEP 05 \(2019\) 043](#) | [arXiv:1812.01691](#) | [DOI:10.1007/JHEP05\(2019\)043](#) | 2018-12-04

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[Phys.Lett.B 796 \(2019\) 131](#) | [arXiv:1812.00380](#) | [DOI:10.1016/j.physletb.2019.07.013](#) | 2018-12-02

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[Eur.Phys.J.C 79 \(2019\) 305](#) | [arXiv:1811.09760](#) | [DOI:10.1140/epjc/s10052-019-6800-x](#) | 2018-11-23

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Sirunyan, Albert M et al.

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Sirunyan, Albert M et al.

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[Eur.Phys.J.C 79 \(2019\) 277](#) | [arXiv:1809.11080](#) | [DOI:10.1140/epjc/s10052-019-6774-8](#) | 2018-09-28

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[Eur.Phys.J.C 79 \(2019\) 421](#) | [arXiv:1809.10733](#) | [DOI:10.1140/epjc/s10052-019-6909-y](#) | 2018-09-27

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[JINST 13 \(2018\) P10034](#) | [arXiv:1807.03289](#) | [DOI:10.1088/1748-0221/13/10/P10034](#) | 2018-07-09

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[Eur.Phys.J.C 79 \(2019\) 20](#) | [arXiv:1807.00782](#) | [DOI:10.1140/epjc/s10052-018-6482-9](#) | 2018-07-02

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[Phys.Lett.B 790 \(2019\) 140](#) | [arXiv:1806.05264](#) | [DOI:10.1016/j.physletb.2019.01.005](#) | 2018-06-13

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[Phys.Rev.Lett. 121 \(2018\) 141802](#) | [arXiv:1806.01058](#) | [DOI:10.1103/PhysRevLett.121.141802](#) | 2018-06-04

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Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 701](#) | [arXiv:1806.00863](#) | [DOI:10.1140/epjc/s10052-018-6148-7](#) | 2018-06-03

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Sirunyan, Albert M et al.

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760. Search for beyond the standard model Higgs bosons decaying into a $b\bar{b}$ pair in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

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Sirunyan, A. M. et al.

[Phys.Rev.Lett. 121 \(2018\) 092002](#) | [arXiv:1805.11192](#) | [DOI:10.1103/PhysRevLett.121.092002](#) | 2018-05-28

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Sirunyan, Albert M et al.

[Phys.Rev.D 98 \(2018\) 032005](#) | [arXiv:1805.10228](#) | [DOI:10.1103/PhysRevD.98.032005](#) | 2018-05-25

763. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state with two b quarks and two τ leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 785 \(2018\) 462](#) | [arXiv:1805.10191](#) | [DOI:10.1016/j.physletb.2018.08.057](#) | 2018-05-25

764. Measurement of nuclear modification factors of $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 790 \(2019\) 270](#) | [arXiv:1805.09215](#) | [DOI:10.1016/j.physletb.2019.01.006](#) | 2018-05-23

765. Measurement of the production cross section for single top quarks in association with W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2018\) 117](#) | [arXiv:1805.07399](#) | [DOI:10.1007/JHEP10\(2018\)117](#) | 2018-05-18

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Sirunyan, Albert M et al.

[JHEP 11 \(2018\) 042](#) | [arXiv:1805.06013](#) | [DOI:10.1007/JHEP11\(2018\)042](#) | 2018-05-15

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Sirunyan, A.M. et al.

[JHEP 09 \(2018\) 065](#) | [arXiv:1805.05784](#) | [DOI:10.1007/JHEP09\(2018\)065](#) | 2018-05-15

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[JHEP 11 \(2018\) 018](#) | [arXiv:1805.04865](#) | [DOI:10.1007/JHEP11\(2018\)018](#) | 2018-05-13

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Sirunyan, Albert M et al.

[Phys.Lett.B 790 \(2019\) 509](#) | [arXiv:1805.02248](#) | [DOI:10.1016/j.physletb.2019.01.058](#) | 2018-05-06

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Sirunyan, A. M. et al.

[Phys.Rev.Lett. 121 \(2018\) 082301](#) | [arXiv:1804.09767](#) | [DOI:10.1103/PhysRevLett.121.082301](#) | 2018-04-25

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Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 016](#) | [arXiv:1804.07321](#) | [DOI:10.1007/JHEP08\(2018\)016](#) | 2018-04-19

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 965](#) | [arXiv:1804.05252](#) | [DOI:10.1140/epjc/s10052-018-6373-0](#) | 2018-04-14

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Sirunyan, A.M. et al.

[JINST 13 \(2018\) P06015](#) | [arXiv:1804.04528](#) | [DOI:10.1088/1748-0221/13/06/P06015](#) | 2018-04-12

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[Phys.Rev.Lett. 120 \(2018\) 231801](#) | [arXiv:1804.02610](#) | [DOI:10.1103/PhysRevLett.120.231801](#) | 2018-04-07

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Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 148](#) | [arXiv:1803.11116](#) | [DOI:10.1007/JHEP05\(2018\)148](#) | 2018-03-29

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784. Search for a heavy resonance decaying into a Z boson and a Z or W boson in $2\ell 2q$ final states at $\sqrt{s}=13$ TeV

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[Phys.Rev.D 97 \(2018\) 112003](#) | [arXiv:1803.08856](#) | [DOI:10.1103/PhysRevD.97.112003](#) | 2018-03-23

786. Search for new physics in dijet angular distributions using proton-proton collisions at $\sqrt{s}= 13$ TeV and constraints on dark matter and other models

Sirunyan, Albert M et al.

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Sirunyan, Albert M et al.

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[JHEP 06 \(2018\) 120](#) | [arXiv:1803.06292](#) | [DOI:10.1007/JHEP06\(2018\)120](#) | 2018-03-16

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[JHEP 08 \(2018\) 066](#) | [arXiv:1803.05485](#) | [DOI:10.1007/JHEP08\(2018\)066](#) | 2018-03-14

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Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 153](#) | [arXiv:1803.04496](#) | [DOI:10.1007/JHEP07\(2018\)153](#) | 2018-03-12

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Sirunyan, A.M. et al.

[JHEP 06 \(2018\) 002](#) | [arXiv:1803.03991](#) | [DOI:10.1007/JHEP06\(2018\)002](#) | 2018-03-11

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Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 075](#) | [arXiv:1803.03838](#) | [DOI:10.1007/JHEP07\(2018\)075](#) | 2018-03-10

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[JHEP 05 \(2018\) 006](#) | [arXiv:1803.00042](#) | [DOI:10.1007/JHEP05\(2018\)006](#) | 2018-02-28

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Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 088](#) | [arXiv:1802.09407](#) | [DOI:10.1007/JHEP05\(2018\)088](#) | 2018-02-26

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[Phys.Rev.Lett. 120 \(2018\) 201801](#) | [arXiv:1802.06149](#) | [DOI:10.1103/PhysRevLett.120.201801](#) | 2018-02-16

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Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 072010](#) | [arXiv:1802.04867](#) | [DOI:10.1103/PhysRevD.97.072010](#) | 2018-02-13

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Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 221801](#) | [arXiv:1802.02965](#) | [DOI:10.1103/PhysRevLett.120.221801](#) | 2018-02-08

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Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 161](#) | [arXiv:1802.02613](#) | [DOI:10.1007/JHEP07\(2018\)161](#) | 2018-02-07

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Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 025](#) | [arXiv:1802.02110](#) | [DOI:10.1007/JHEP05\(2018\)025](#) | 2018-02-06

802. Search for single production of vector-like quarks decaying to a b quark and a Higgs boson

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[JHEP 06 \(2018\) 031](#) | [arXiv:1802.01486](#) | [DOI:10.1007/JHEP06\(2018\)031](#) | 2018-02-05

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Sirunyan, Albert M et al.

[JHEP 04 \(2018\) 073](#) | [arXiv:1802.01122](#) | [DOI:10.1007/JHEP04\(2018\)073](#) | 2018-02-04

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Sirunyan, Albert M et al.

[JHEP 03 \(2018\) 181](#) | [arXiv:1802.00707](#) | [DOI:10.1007/JHEP03\(2018\)181](#) | 2018-02-02

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[Phys.Rev.Lett. 121 \(2018\) 242301](#) | [arXiv:1801.04895](#) | [DOI:10.1103/PhysRevLett.121.242301](#) | 2018-01-15

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[Eur.Phys.J.C 78 \(2018\) 708](#) | [arXiv:1801.03535](#) | [DOI:10.1140/epjc/s10052-018-6146-9](#) | 2018-01-10

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[Phys.Lett.B 782 \(2018\) 440](#) | [arXiv:1801.01846](#) | [DOI:10.1016/j.physletb.2018.05.062](#) | 2018-01-05

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810. Search for R-parity violating supersymmetry in pp collisions at $\sqrt{s} = 13$ TeV using b jets in a final state with a single lepton, many jets, and high sum of large-radius jet masses

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[Phys.Lett.B 783 \(2018\) 114](#) | [arXiv:1712.08920](#) | [DOI:10.1016/j.physletb.2018.06.028](#) | 2017-12-24

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Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 509](#) | [arXiv:1712.08959](#) | [DOI:10.1140/epjc/s10052-018-5950-6](#) | 2017-12-24

812. Search for Physics Beyond the Standard Model in Events with High-Momentum Higgs Bosons and Missing Transverse Momentum in Proton-Proton Collisions at 13 TeV

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[Phys.Rev.Lett. 120 \(2018\) 241801](#) | [arXiv:1712.08501](#) | [DOI:10.1103/PhysRevLett.120.241801](#) | 2017-12-22

813. Search for lepton flavour violating decays of the Higgs boson to $\mu\tau$ and $e\tau$ in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 06 \(2018\) 001](#) | [arXiv:1712.07173](#) | [DOI:10.1007/JHEP06\(2018\)001](#) | 2017-12-19

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[Phys.Rev.C 97 \(2018\) 064912](#) | [arXiv:1712.07198](#) | [DOI:10.1103/PhysRevC.97.064912](#) | 2017-12-19

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[JINST 13 \(2018\) P05011](#) | [arXiv:1712.07158](#) | [DOI:10.1088/1748-0221/13/05/P05011](#) | 2017-12-19

816. Search for the X(5568) state decaying into $B^0\pi^\pm$ in proton-proton collisions at $\sqrt{s}=8$ TeV

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[Phys.Rev.Lett. 120 \(2018\) 202005](#) | [arXiv:1712.06144](#) | [DOI:10.1103/PhysRevLett.120.202005](#) | 2017-12-17

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[Eur.Phys.J.C 78 \(2018\) 566](#) | [arXiv:1712.05471](#) | [DOI:10.1140/epjc/s10052-018-6033-4](#) | 2017-12-14

818. Search for $Z\gamma$ resonances using leptonic and hadronic final states in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 09 \(2018\) 148](#) | [arXiv:1712.03143](#) | [DOI:10.1007/JHEP09\(2018\)148](#) | 2017-12-08

819. Measurement of the associated production of a single top quark and a Z boson in pp collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 779 \(2018\) 358](#) | [arXiv:1712.02825](#) | [DOI:10.1016/j.physletb.2018.02.025](#) | 2017-12-07

820. Search for the flavor-changing neutral current interactions of the top quark and the Higgs boson which decays into a pair of b quarks at $\sqrt{s}=13$ TeV

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[JHEP 06 \(2018\) 102](#) | [arXiv:1712.02399](#) | [DOI:10.1007/JHEP06\(2018\)102](#) | 2017-12-06

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Sirunyan, Albert M et al.

[JHEP 02 \(2018\) 032](#) | [arXiv:1712.02280](#) | [DOI:10.1007/JHEP02\(2018\)032](#) | 2017-12-06

822. Search for new physics in final states with an energetic jet or a hadronically decaying W or Z boson and transverse momentum imbalance at $\sqrt{s}=13$ TeV

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[Phys.Rev.D 97 \(2018\) 092005](#) | [arXiv:1712.02345](#) | [DOI:10.1103/PhysRevD.97.092005](#) | 2017-12-06

823. Search for pair production of excited top quarks in the lepton + jets final state

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[Phys.Lett.B 778 \(2018\) 349](#) | [arXiv:1711.10949](#) | [DOI:10.1016/j.physletb.2018.01.049](#) | 2017-11-29

824. Study of jet quenching with isolated-photon+jet correlations in PbPb and pp collisions at $\sqrt{s_{NN}}=5.02$ TeV

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[Phys.Lett.B 785 \(2018\) 14](#) | [arXiv:1711.09738](#) | [DOI:10.1016/j.physletb.2018.07.061](#) | 2017-11-27

825. Search for new long-lived particles at $\sqrt{s} = 13$ TeV

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[Phys.Lett.B 780 \(2018\) 432](#) | [arXiv:1711.09120](#) | [DOI:10.1016/j.physletb.2018.03.019](#) | 2017-11-24

826. Search for gauge-mediated supersymmetry in events with at least one photon and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV

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[Phys.Lett.B 780 \(2018\) 118](#) | [arXiv:1711.08008](#) | [DOI:10.1016/j.physletb.2018.02.045](#) | 2017-11-21

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[Phys.Lett.B 789 \(2019\) 643](#) | [arXiv:1711.05594](#) | [DOI:10.1016/j.physletb.2018.11.063](#) | 2017-11-15

828. Search for excited quarks of light and heavy flavor in γ + jet final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Lett.B 781 \(2018\) 390](#) | [arXiv:1711.04652](#) | [DOI:10.1016/j.physletb.2018.04.007](#) | 2017-11-13

829. Search for ZZ resonances in the $2\ell 2\nu$ final state in proton-proton collisions at 13 TeV

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[JHEP 03 \(2018\) 003](#) | [arXiv:1711.04370](#) | [DOI:10.1007/JHEP03\(2018\)003](#) | 2017-11-12

830. Measurement of the underlying event activity in inclusive Z boson production in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 07 \(2018\) 032](#) | [arXiv:1711.04299](#) | [DOI:10.1007/JHEP07\(2018\)032](#) | 2017-11-12

831. Measurement of the inclusive tt cross section in pp collisions at $\sqrt{s}=5.02$ TeV using final states with at least one charged lepton

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[JHEP 03 \(2018\) 115](#) | [arXiv:1711.03143](#) | [DOI:10.1007/JHEP03\(2018\)115](#) | 2017-11-08

832. Measurement of the cross section for top quark pair production in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 011](#) | [arXiv:1711.02547](#) | [DOI:10.1007/JHEP08\(2018\)011](#) | 2017-11-07

833. Measurement of associated Z + charm production in proton-proton collisions at $\sqrt{s} = 8$ TeV

Sirunyan, A.M. et al.

[Eur.Phys.J.C 78 \(2018\) 287](#) | [arXiv:1711.02143](#) | [DOI:10.1140/epjc/s10052-018-5752-x](#) | 2017-11-06

834. Search for top squarks and dark matter particles in opposite-charge dilepton final states at $\sqrt{s}= 13$ TeV

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[Phys.Rev.D 97 \(2018\) 032009](#) | [arXiv:1711.00752](#) | [DOI:10.1103/PhysRevD.97.032009](#) | 2017-11-02

835. Search for new physics in events with a leptonically decaying Z boson and a large transverse momentum imbalance in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 78 \(2018\) 291](#) | [arXiv:1711.00431](#) | [DOI:10.1140/epjc/s10052-018-5740-1](#) | 2017-11-01

836. Measurement of quarkonium production cross sections in pp collisions at $\sqrt{s}= 13$ TeV

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[Phys.Lett.B 780 \(2018\) 251](#) | [arXiv:1710.11002](#) | [DOI:10.1016/j.physletb.2018.02.033](#) | 2017-10-30

837. Search for supersymmetry in proton-proton collisions at 13 TeV using identified top quarks

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[Phys.Rev.D 97 \(2018\) 012007](#) | [arXiv:1710.11188](#) | [DOI:10.1103/PhysRevD.97.012007](#) | 2017-10-30

838. Search for supersymmetry in events with at least three electrons or muons, jets, and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 02 \(2018\) 067](#) | [arXiv:1710.09154](#) | [DOI:10.1007/JHEP02\(2018\)067](#) | 2017-10-25

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[JHEP 01 \(2018\) 045](#) | [arXiv:1710.09355](#) | [DOI:10.1007/JHEP01\(2018\)045](#) | 2017-10-25

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[Phys.Lett.B 780 \(2018\) 501](#) | [arXiv:1709.07497](#) | [DOI:10.1016/j.physletb.2018.02.050](#) | 2017-09-21

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[Phys.Lett.B 778 \(2018\) 101](#) | [arXiv:1707.02909](#) | [DOI:10.1016/j.physletb.2018.01.001](#) | 2017-07-10

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[Phys.Lett.B 775 \(2017\) 315](#) | [arXiv:1706.08578](#) | [DOI:10.1016/j.physletb.2017.11.001](#) | 2017-06-26

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Khachatryan, Vardan et al.

[JHEP 09 \(2016\) 051](#) | [arXiv:1605.02329](#) | [DOI:10.1007/JHEP09\(2016\)051](#) | 2016-05-08

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Khachatryan, Vardan et al.

[Phys.Rev.D 94 \(2016\) 072002](#) | [arXiv:1605.00116](#) | [DOI:10.1103/PhysRevD.94.072002](#) | 2016-04-30

1007. Search for narrow resonances in dijet final states at $\sqrt{s}=8$ TeV with the novel CMS technique of data scouting

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 117 \(2016\) 031802](#) | [arXiv:1604.08907](#) | [DOI:10.1103/PhysRevLett.117.031802](#) | 2016-04-29

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Khachatryan, Vardan et al.

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[Phys.Rev.C 96 \(2017\) 014915](#) | [arXiv:1604.05347](#) | [DOI:10.1103/PhysRevC.96.014915](#) | 2016-04-18

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Khachatryan, Vardan et al.

[JHEP 08 \(2016\) 119](#) | [arXiv:1604.04464](#) | [DOI:10.1007/JHEP08\(2016\)119](#) | 2016-04-15

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[JHEP 12 \(2016\) 088](#) | [arXiv:1603.08914](#) | [DOI:10.1007/JHEP12\(2016\)088](#) | 2016-03-29

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[Phys.Rev.D 94 \(2016\) 052012](#) | [arXiv:1603.06896](#) | [DOI:10.1103/PhysRevD.94.052012](#) | 2016-03-22

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Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 092006](#) | [arXiv:1603.06536](#) | [DOI:10.1103/PhysRevD.93.092006](#) | 2016-03-21

1014. Search for neutral resonances decaying into a Z boson and a pair of b jets or τ leptons

Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 369](#) | [arXiv:1603.02991](#) | [DOI:10.1016/j.physletb.2016.05.087](#) | 2016-03-09

1015. $\Upsilon(nS)$ polarizations versus particle multiplicity in pp collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 761 \(2016\) 31](#) | [arXiv:1603.02913](#) | [DOI:10.1016/j.physletb.2016.07.065](#) | 2016-03-09

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Khachatryan, Vardan et al.

[JHEP 09 \(2016\) 027](#) | [arXiv:1603.02555](#) | [DOI:10.1007/JHEP09\(2016\)027](#) | 2016-03-08

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Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 169](#) | [arXiv:1603.02248](#) | [DOI:10.1007/JHEP04\(2016\)169](#) | 2016-03-07

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Khachatryan, Vardan et al.

[JHEP 08 \(2016\) 029](#) | [arXiv:1603.02303](#) | [DOI:10.1007/JHEP08\(2016\)029](#) | 2016-03-07

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Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 469](#) | [arXiv:1603.01803](#) | [DOI:10.1140/epjc/s10052-016-4293-4](#) | 2016-03-06

1020. Search for direct pair production of supersymmetric top quarks decaying to all-hadronic final states in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 460](#) | [arXiv:1603.00765](#) | [DOI:10.1140/epjc/s10052-016-4292-5](#) | 2016-03-02

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Khachatryan, Vardan et al.

[Eur.Phys.J.C 77 \(2017\) 15](#) | [arXiv:1602.09024](#) | [DOI:10.1140/epjc/s10052-016-4504-z](#) | 2016-02-29

1022. Search for heavy resonances decaying to two Higgs bosons in final states containing four b quarks

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 371](#) | [arXiv:1602.08762](#) | [DOI:10.1140/epjc/s10052-016-4206-6](#) | 2016-02-28

1023. Search for supersymmetry in electroweak production with photons and large missing transverse energy in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 479](#) | [arXiv:1602.08772](#) | [DOI:10.1016/j.physletb.2016.05.088](#) | 2016-02-28

1024. Measurement of the $Z \rightarrow \nu \bar{\nu} \gamma$ production cross section in pp collisions at $\sqrt{s}=8$ TeV and limits on anomalous ZZ γ and Z $\gamma\gamma$ trilinear gauge boson couplings

Khachatryan, Vardan et al.

[Phys.Lett.B 760 \(2016\) 448](#) | [arXiv:1602.07152](#) | [DOI:10.1016/j.physletb.2016.06.080](#) | 2016-02-23

1025. Search for supersymmetry in the multijet and missing transverse momentum final state in pp collisions at 13 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 758 \(2016\) 152](#) | [arXiv:1602.06581](#) | [DOI:10.1016/j.physletb.2016.05.002](#) | 2016-02-21

1026. Measurement of dijet azimuthal decorrelation in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 536](#) | [arXiv:1602.04384](#) | [DOI:10.1140/epjc/s10052-016-4346-8](#) | 2016-02-13

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[Phys.Lett.B 760 \(2016\) 178](#) | [arXiv:1602.04334](#) | [DOI:10.1016/j.physletb.2016.06.039](#) | 2016-02-13

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[Phys.Rev.D 93 \(2016\) 092009](#) | [arXiv:1602.02917](#) | [DOI:10.1103/PhysRevD.93.092009](#) | 2016-02-09

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Khachatryan, Vardan et al.

[JHEP 07 \(2016\) 027](#) | [arXiv:1602.03169](#) | [DOI:10.1007/JHEP07\(2016\)027](#) | 2016-02-09

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Khachatryan, Vardan et al.

[JHEP 08 \(2016\) 139](#) | [arXiv:1601.06713](#) | [DOI:10.1007/JHEP08\(2016\)139](#) | 2016-01-25

1032. Search for massive WH resonances decaying into the $\ell \nu b \bar{b}$ final state at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 237](#) | [arXiv:1601.06431](#) | [DOI:10.1140/epjc/s10052-016-4067-z](#) | 2016-01-24

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Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 325](#) | [arXiv:1601.04768](#) | [DOI:10.1140/epjc/s10052-016-4156-z](#) | 2016-01-18

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[Eur.Phys.J.C 76 \(2016\) 372](#) | [arXiv:1601.02001](#) | [DOI:10.1140/epjc/s10052-016-4205-7](#) | 2016-01-08

1035. Correlations between jets and charged particles in PbPb and pp collisions at $\sqrt{s_{NN}}=2.76$ TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2016\) 156](#) | [arXiv:1601.00079](#) | [DOI:10.1007/JHEP02\(2016\)156](#) | 2016-01-01

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[JHEP 04 \(2016\) 005](#) | [arXiv:1512.08377](#) | [DOI:10.1007/JHEP04\(2016\)005](#) | 2015-12-28

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[Phys.Lett.B 759 \(2016\) 9](#) | [arXiv:1512.08002](#) | [DOI:10.1016/j.physletb.2016.05.033](#) | 2015-12-25

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[Phys.Lett.B 759 \(2016\) 36](#) | [arXiv:1512.06461](#) | [DOI:10.1016/j.physletb.2016.05.044](#) | 2015-12-20

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Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 155](#) | [arXiv:1512.00815](#) | [DOI:10.1140/epjc/s10052-016-3988-x](#) | 2015-12-02

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Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 052011](#) | [arXiv:1511.09375](#) | [DOI:10.1103/PhysRevD.93.052011](#) | 2015-11-30

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[Phys.Lett.B 758 \(2016\) 321](#) | [arXiv:1511.06170](#) | [DOI:10.1016/j.physletb.2016.05.005](#) | 2015-11-19

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

[Phys.Lett.B 758 \(2016\) 296](#) | [arXiv:1511.03610](#) | [DOI:10.1016/j.physletb.2016.05.003](#) | 2015-11-11

1046. Measurement of Top Quark Polarisation in T-Channel Single Top Quark Production

Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 073](#) | [arXiv:1511.02138](#) | [DOI:10.1007/JHEP04\(2016\)073](#) | 2015-11-06

1047. Search for Excited Leptons in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2016\) 125](#) | [arXiv:1511.01407](#) | [DOI:10.1007/JHEP03\(2016\)125](#) | 2015-11-04

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

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1053. Measurement of long-range near-side two-particle angular correlations in pp collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 116 \(2016\) 172302](#) | [arXiv:1510.03068](#) | [DOI:10.1103/PhysRevLett.116.172302](#) | 2015-10-11

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Khachatryan, Vardan et al.

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[Phys.Lett.B 755 \(2016\) 217](#) | [arXiv:1510.01181](#) | [DOI:10.1016/j.physletb.2016.01.056](#) | 2015-10-05

1057. Measurement of transverse momentum relative to dijet systems in PbPb and pp collisions at $\sqrt{s_{NN}}=2.76$ TeV

Khachatryan, Vardan et al.

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1058. Search for the production of an excited bottom quark decaying to tW in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2016\) 166](#) | [arXiv:1509.08141](#) | [DOI:10.1007/JHEP01\(2016\)166](#) | 2015-09-27

1059. Search for the associated production of a Higgs boson with a single top quark in proton-proton collisions at $\sqrt{s}=8$ TeV

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[JHEP 06 \(2016\) 177](#) | [arXiv:1509.08159](#) | [DOI:10.1007/JHEP06\(2016\)177](#) | 2015-09-27

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[Phys.Lett.B 759 \(2016\) 641](#) | [arXiv:1509.03893](#) | [DOI:10.1016/j.physletb.2016.06.027](#) | 2015-09-13

1065. Search for pair production of first and second generation leptoquarks in proton-proton collisions at $\sqrt{s} = 8$ TeV

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[Phys.Rev.D 93 \(2016\) 032004](#) | [arXiv:1509.03744](#) | [DOI:10.1103/PhysRevD.93.032004](#) | 2015-09-12

1066. Search for single production of scalar leptoquarks in proton-proton collisions at $\sqrt{s} = 8$ TeV

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[Phys.Rev.D 93 \(2016\) 032005](#) | [arXiv:1509.03750](#) | [DOI:10.1103/PhysRevD.93.032005](#) | 2015-09-12

1067. Search for a charged Higgs boson in pp collisions at $\sqrt{s} = 8$ TeV

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[JHEP 11 \(2015\) 018](#) | [arXiv:1508.07774](#) | [DOI:10.1007/JHEP11\(2015\)018](#) | 2015-08-31

1068. Measurement of differential cross sections for Higgs boson production in the diphoton decay channel in pp collisions at $\sqrt{s} = 8$ TeV

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[Eur.Phys.J.C 76 \(2016\) 13](#) | [arXiv:1508.07819](#) | [DOI:10.1140/epjc/s10052-015-3853-3](#) | 2015-08-31

1069. Search for supersymmetry in the vector-boson fusion topology in proton-proton collisions at $\sqrt{s} = 8$ TeV

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[JHEP 11 \(2015\) 189](#) | [arXiv:1508.07628](#) | [DOI:10.1007/JHEP11\(2015\)189](#) | 2015-08-30

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[Phys.Rev.Lett. 116 \(2016\) 032301](#) | [arXiv:1508.06678](#) | [DOI:10.1103/PhysRevLett.116.032301](#) | 2015-08-26

1071. Search for W' decaying to tau lepton and neutrino in proton-proton collisions at $\sqrt{s} = 8$ TeV

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[Phys.Lett.B 755 \(2016\) 196](#) | [arXiv:1508.04308](#) | [DOI:10.1016/j.physletb.2016.02.002](#) | 2015-08-18

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[Phys.Rev.D 93 \(2016\) 034014](#) | [arXiv:1508.03862](#) | [DOI:10.1103/PhysRevD.93.034014](#) | 2015-08-16

1073. Search for neutral MSSM Higgs bosons decaying to $\mu^+\mu^-$ in pp collisions at $\sqrt{s} = 7$ and 8 TeV

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[Phys.Lett.B 752 \(2016\) 221](#) | [arXiv:1508.01437](#) | [DOI:10.1016/j.physletb.2015.11.042](#) | 2015-08-06

1074. Search for supersymmetry in events with a photon, a lepton, and missing transverse momentum in pp collisions at $\sqrt{s} = 8$ TeV

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[Phys.Lett.B 757 \(2016\) 6](#) | [arXiv:1508.01218](#) | [DOI:10.1016/j.physletb.2016.03.039](#) | 2015-08-05

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[Phys.Lett.B 753 \(2016\) 424](#) | [arXiv:1507.08126](#) | [DOI:10.1016/j.physletb.2015.12.020](#) | 2015-07-29

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[Phys.Rev.D 92 \(2015\) 072010](#) | [arXiv:1507.06656](#) | [DOI:10.1103/PhysRevD.92.072010](#) | 2015-07-23

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[Phys.Lett.B 751 \(2015\) 143](#) | [arXiv:1507.05915](#) | [DOI:10.1016/j.physletb.2015.10.004](#) | 2015-07-21

1081. Inclusive and differential measurements of the tt charge asymmetry in pp collisions at $\sqrt{s} = 8$ TeV

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[Phys.Lett.B 757 \(2016\) 154](#) | [arXiv:1507.03119](#) | [DOI:10.1016/j.physletb.2016.03.060](#) | 2015-07-11

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[Phys.Rev.D 92 \(2015\) 072006](#) | [arXiv:1507.02898](#) | [DOI:10.1103/PhysRevD.92.072006](#) | 2015-07-10

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[Phys.Lett.B 753 \(2016\) 341](#) | [arXiv:1507.03031](#) | [DOI:10.1016/j.physletb.2015.12.039](#) | 2015-07-10

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[Phys.Rev.D 92 \(2015\) 112001](#) | [arXiv:1507.00233](#) | [DOI:10.1103/PhysRevD.92.112001](#) | 2015-07-01

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Khachatryan, Vardan et al.

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[Phys.Rev.D 92 \(2015\) 012003](#) | [arXiv:1503.08689](#) | [DOI:10.1103/PhysRevD.92.012003](#) | 2015-03-30

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[Phys.Rev.Lett. 114 \(2015\) 191803](#) | [arXiv:1503.07589](#) | [DOI:10.1103/PhysRevLett.114.191803](#) | 2015-03-26

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[Phys.Lett.B 750 \(2015\) 565](#) | [arXiv:1503.05825](#) | [DOI:10.1016/j.physletb.2015.09.057](#) | 2015-03-19

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[Phys.Lett.B 749 \(2015\) 560](#) | [arXiv:1503.04114](#) | [DOI:10.1016/j.physletb.2015.08.047](#) | 2015-03-13

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[Eur.Phys.J.C 75 \(2015\) 237](#) | [arXiv:1502.05387](#) | [DOI:10.1140/epjc/s10052-015-3435-4](#) | 2015-02-18

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[Phys.Rev.Lett. 115 \(2015\) 012301](#) | [arXiv:1502.05382](#) | [DOI:10.1103/PhysRevLett.115.012301](#) | 2015-02-18

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Khachatryan, Vardan et al.

[Phys.Lett.B 748 \(2015\) 255](#) | [arXiv:1502.04994](#) | [DOI:10.1016/j.physletb.2015.07.011](#) | 2015-02-17

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[Eur.Phys.J.C 75 \(2015\) 302](#) | [arXiv:1502.04785](#) | [DOI:10.1140/epjc/s10052-015-3491-9](#) | 2015-02-16

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[Eur.Phys.J.C 75 \(2015\) 251](#) | [arXiv:1502.02485](#) | [DOI:10.1140/epjc/s10052-015-3454-1](#) | 2015-02-09

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[Phys.Lett.B 749 \(2015\) 14](#) | [arXiv:1501.07750](#) | [DOI:10.1016/j.physletb.2015.07.037](#) | 2015-01-30

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[Phys.Lett.B 756 \(2016\) 84](#) | [arXiv:1501.06089](#) | [DOI:10.1016/j.physletb.2016.02.047](#) | 2015-01-24

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[Eur.Phys.J.C 75 \(2015\) 151](#) | [arXiv:1501.05603](#) | [DOI:10.1140/epjc/s10052-015-3367-z](#) | 2015-01-22

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[Phys.Lett.B 748 \(2015\) 144](#) | [arXiv:1501.05566](#) | [DOI:10.1016/j.physletb.2015.06.070](#) | 2015-01-22

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[Phys.Lett.B 745 \(2015\) 5](#) | [arXiv:1412.4109](#) | [DOI:10.1016/j.physletb.2015.04.002](#) | 2014-12-12

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[Phys.Lett.B 743 \(2015\) 503](#) | [arXiv:1411.7255](#) | [DOI:10.1016/j.physletb.2015.03.017](#) | 2014-11-26

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1140. Search for quark contact interactions and extra spatial dimensions using dijet angular distributions in proton-proton collisions at $\sqrt{s} = 8$ TeV

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[Phys.Lett.B 746 \(2015\) 79](#) | [arXiv:1411.2646](#) | [DOI:10.1016/j.physletb.2015.04.042](#) | 2014-11-10

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[JINST 10 \(2015\) P02006](#) | [arXiv:1411.0511](#) | [DOI:10.1088/1748-0221/10/02/P02006](#) | 2014-11-03

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[Phys.Lett.B 755 \(2016\) 102](#) | [arXiv:1410.8812](#) | [DOI:10.1016/j.physletb.2016.01.057](#) | 2014-10-31

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[Phys.Lett.B 744 \(2015\) 184](#) | [arXiv:1410.6679](#) | [DOI:10.1016/j.physletb.2015.03.048](#) | 2014-10-24

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[Eur.Phys.J.C 75 \(2015\) 288](#) | [arXiv:1410.6765](#) | [DOI:10.1140/epjc/s10052-015-3499-1](#) | 2014-10-24

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[Phys.Rev.D 90 \(2014\) 112013](#) | [arXiv:1410.2751](#) | [DOI:10.1103/PhysRevD.90.112013](#) | 2014-10-10

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[Phys.Rev.Lett. 113 \(2014\) 262301](#) | [arXiv:1410.1804](#) | [DOI:10.1103/PhysRevLett.113.262301](#) | 2014-10-07

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[Phys.Lett.B 742 \(2015\) 200](#) | [arXiv:1409.3392](#) | [DOI:10.1016/j.physletb.2015.01.034](#) | 2014-09-11

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1158. Search for neutral MSSM Higgs bosons decaying to a pair of tau leptons in pp collisions

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[Phys.Rev.D 91 \(2015\) 052008](#) | [arXiv:1408.3104](#) | [DOI:10.1103/PhysRevD.91.052008](#) | 2014-08-13

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[JHEP 09 \(2014\) 087](#) | [arXiv:1408.1682](#) | [DOI:10.1007/JHEP09\(2014\)087](#) | 2014-08-07

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[Eur.Phys.J.C 74 \(2014\) 3149](#) | [arXiv:1407.3683](#) | [DOI:10.1140/epjc/s10052-014-3149-z](#) | 2014-07-14

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[Phys.Lett.B 740 \(2015\) 83](#) | [arXiv:1407.3476](#) | [DOI:10.1016/j.physletb.2014.11.026](#) | 2014-07-13

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[Phys.Lett.B 731 \(2014\) 173](#) | [arXiv:1310.1618](#) | [DOI:10.1016/j.physletb.2014.02.033](#) | 2013-10-06

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[JINST 8 \(2013\) P11002](#) | [arXiv:1306.6905](#) | [DOI:10.1088/1748-0221/8/11/P11002](#) | 2013-06-28

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